

CONSOLIDATED GUIDE

Implementation Acknowledgment For Health Care Insurance (999)

Consolidated Documents:

June 2007 005010X231

June 2010 005010X231A1

February 2011

ASC X12 Consolidated Guides aid in transaction implementation by combining official material into one user friendly document. Although the consolidated guides have not been explicitly mandated under HIPAA, they incorporate the individual guides that have been named into single documents. In the event that there is a conflict between the Consolidated Guides and the ASC X12 Type 3 Technical Reports or any subsequent errata, the underlying ASC X12 publications are the authoritative source.

Copyright © 2011, Data Interchange Standards Association (DISA) on behalf of the membership of ASC X12. Format © 2011 Washington Publishing Company (WPC). Exclusively published by WPC under license from DISA. All Rights Reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without prior written permission.

Table of Contents

1 Purpose and Business Information	5
1.1 Implementation Purpose and Scope	5
1.2 Version Information	5
1.3 Implementation Limitations	5
1.3.1 Batch and Real-time Usage	5
1.3.2 Other Usage Limitations	6
1.4 Business Usage	6
1.5 Business Terminology	6
1.6 Transaction Acknowledgments	7
1.7 Related Transactions	7
1.8 Trading Partner Agreements	7
1.9 Data Overview	7
1.9.1 Overall Data Architecture	7
1.9.1.1 Response Process	7
2 Transaction Sets	9
2.1 Presentation Examples	9
2.2 Implementation Usage	13
2.2.1 Industry Usage	13
2.2.1.1 Transaction Compliance Related to Industry Usage	14
2.2.2 Loops	14
2.3 Transaction Set Listing	15
2.3.1 Implementation	15
2.3.2 X12 Standard	17
2.4 Segment Detail	20
ST Transaction Set Header	21
AK1 Functional Group Response Header	23
AK2 Transaction Set Response Header	25
IK3 Error Identification	27
CTX Segment Context	29
CTX Business Unit Identifier	32
IK4 Implementation Data Element Note	34
CTX Element Context	36
IK5 Transaction Set Response Trailer	39
AK9 Functional Group Response Trailer	42
SE Transaction Set Trailer	45
3 Example	47
3.1 EDI Transmission Example	47

A	External Code Sources	A.1
	77 X12 Directories	A.1
	881 Version / Release / Industry Identifier Code	A.1
B	Nomenclature	B.1
B.1	ASC X12 Nomenclature	B.1
	B.1.1 Interchange and Application Control Structures	B.1
	B.1.1.1 Interchange Control Structure	B.1
	B.1.1.2 Application Control Structure Definitions and Concepts	B.2
	B.1.1.3 Business Transaction Structure Definitions and Concepts	B.5
	B.1.1.4 Envelopes and Control Structures	B.13
	B.1.1.5 Acknowledgments	B.16
B.2	Object Descriptors	B.16
C	EDI Control Directory Including Implementation Usage	C.1
C.1	Control Segments	C.1
	ISA Interchange Control Header	C.3
	TA1 Interchange Acknowledgment Header	C.10
	GS Functional Group Header	C.12
	GE Functional Group Trailer	C.15
	IEA Interchange Control Trailer	C.16
D	Change Summary	D.1
E	Data Element Glossary	E.1
E.1	Data Element Name Index	E.1

1 Purpose and Business Information

1.1 Implementation Purpose and Scope

The purpose of this implementation guide is to provide standardized data content and structure to users of the ASC X12 999 transaction set for Health Care Insurance. This implementation guide is intended to enable a receiver of a functional group based on an X12 Implementation Guideline (TR3) related to Health Care Insurance business processes, to report the syntactical and relational analysis as specified by that implementation guideline (TR3), or to acknowledge receipt of an error-free transaction set.

This 999 is not limited to only Implementation Guide (IG) errors. It can report standard syntax errors, as well as IG errors.

This 999 implementation guide can NOT be used for any application level validations.

1.2 Version Information

This implementation guide is based on the October 2003 ASC X12 standards, referred to as Version 5, Release 1, Sub-release 0 (005010). The unique Version/Release/Industry Identifier Code for transaction sets that are defined by this implementation guide is 005010**X231A1**.

The two-character Functional Identifier Code for the transaction set included in this implementation guide:

- **FA** *Functional or Implementation Acknowledgment Transaction Sets (997, 999)*

The Version/Release/Industry Identifier Code and the applicable Functional Identifier Code must be transmitted in the Functional Group Header (GS segment) that begins a functional group of these transaction sets. For more information, see the descriptions of GS01 and GS08 in Appendix C.

1.3 Implementation Limitations

1.3.1 Batch and Real-Time Usage

There are multiple methods available for sending and receiving business transactions electronically. Two common modes for EDI transactions are batch and real-time.

Batch - In a batch mode the sender does not remain connected while the receiver processes the transactions. Processing is usually completed according to a set schedule. If there is an associated business response transaction (such as a 271 Response to a 270 Request for Eligibility), the receiver creates the response transaction and stores it for future delivery. The sender of the original transmission reconnects at a later time and picks up the response transaction.

This implementation guide does not set specific response time parameters for these activities.

Real Time - In real-time mode the sender remains connected while the receiver processes the transactions and returns a response transaction to the sender. This implementation guide does not set specific response time parameters for implementers.

This implementation guide is intended to support use in batch mode. This implementation guide is intended to support use in real-time mode. A statement that the transaction is not intended to support a specific mode does not preclude its use in that mode between willing trading partners.

1.3.2 Other Usage Limitations

The ASC X12 999 transaction set is designed to report only on conformance against an implementation guideline (TR3).

The 999 is not limited to only IG errors. It can report standard syntax errors, as well as IG errors.

This 999 implementation guide can NOT be used for any application level validations.

The ASC X12 999 transaction set is designed to respond to one and only one functional group (i.e. GS/GE), but will respond to all transaction sets (i.e. ST/SE) within that functional group.

When acknowledging a healthcare implementation guide (TR3) based transaction, only a single transaction set 999 can be used. When acknowledging a healthcare implementation guide (TR3) based transaction, the transaction set 997 must not be used.

This ASC X12 999 Implementation Guideline can NOT be used to respond to any management transaction sets intended for acknowledgements, i.e. TS 997 and 999, or interchange control segments related to acknowledgments, i.e. TA1 and TA3.

1.4 Business Usage

This ASC X12 999 implementation guide (TR3) is intended to meet the needs of the Health Care industry as a whole, for a standard implementation guideline designed for reporting of syntactical errors against a functional group based on an X12 Implementation Guideline, or to report receipt of a functional group that fully complies with an implementation guideline.

For more information on the relationship between the 999 transaction set and other response transaction sets, refer to the ASC X12 document "Reference Model for the Acknowledgment and Tracking of EDI Interchanges".

1.5 Business Terminology

No special business terms are used in this implementation guide.

1.6 Transaction Acknowledgments

This is one of several acknowledgment transactions available for use. Acknowledgment transactions may be used at the discretion of the trading partners.

1.7 Related Transactions

This ASC X12 999 Implementation Guide (TR3) is designed for responding to Health Care transactions based upon an Implementation Guide, including, but not limited to, the following:

269 005010X187
270 005010X279
271 005010X279
274 005030X209
275 005040X254
276 005010X212
277 005010X212
278 005010X217
820 005010X218
834 005010X220
835 005010X221
837 005010X222
837 005010X223
837 005010X224

1.8 Trading Partner Agreements

Trading partner agreements are used to establish and document the relationship between trading partners. A trading partner agreement must not override the specifications in this implementation guide if a transmission is reported in GS08 to be a product of this implementation guide.

1.9 Data Overview

1.9.1 Overall Data Architecture

NOTE

See Appendix B, Nomenclature, to review the transaction set structure, including descriptions of segments, data elements, levels and loops.

1.9.1.1 Response Process

The following sequence diagram (sd) shows how the 999 transaction set is used with other X12 response transactions.

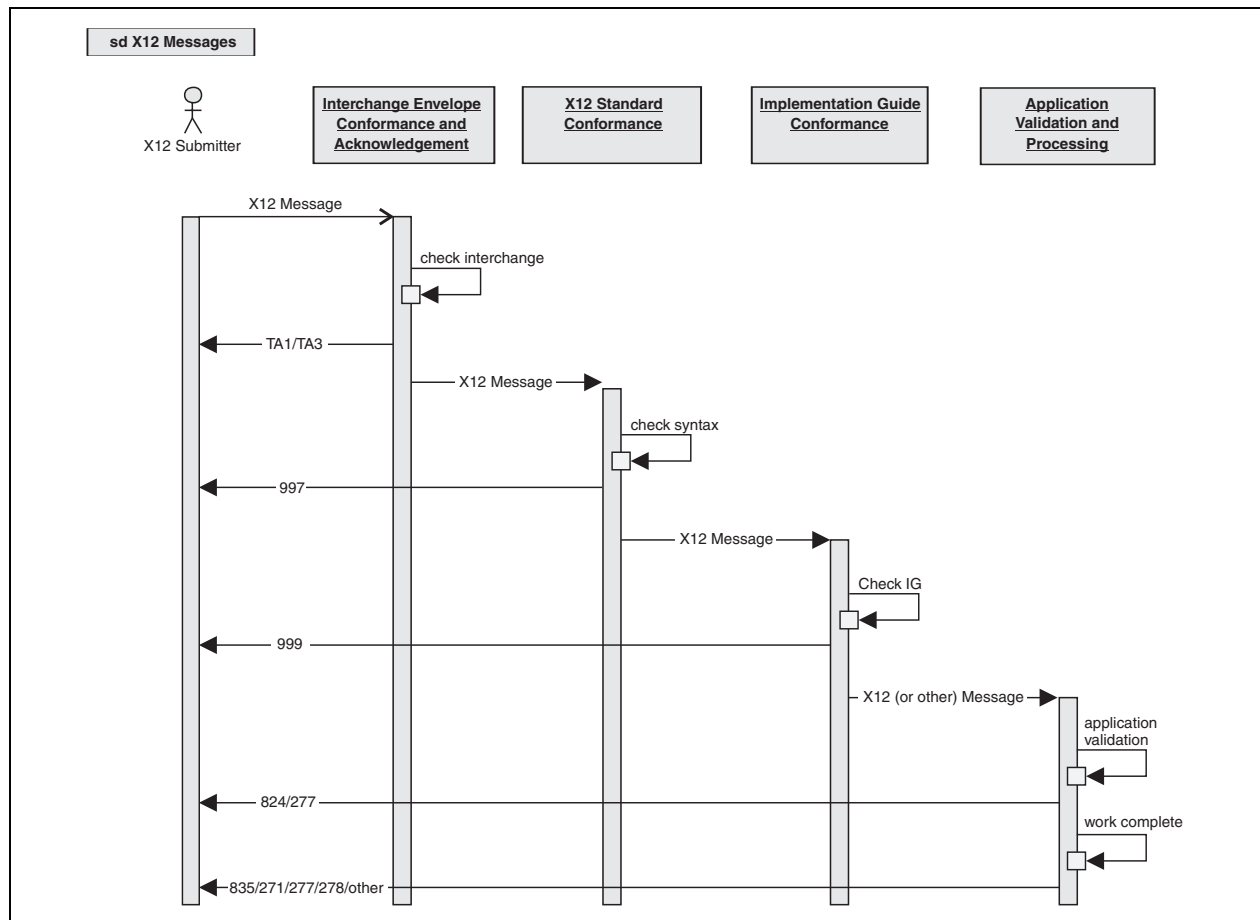


Figure 1.1. Sequence Diagram

For more information on the relationship between the 999 transaction set and other response transaction sets, refer to the ASC X12 document “Reference Model for the Acknowledgement and Tracking of EDI Interchanges”.

2 Transaction Set

NOTE

See Appendix B, Nomenclature, to review the transaction set structure, including descriptions of segments, data elements, levels, and loops.

2.1 Presentation Examples

The ASC X12 standards are generic. For example, multiple trading communities use the same PER segment to specify administrative communication contacts. Each community decides which elements to use and which code values in those elements are applicable.

In this implementation guide, **IMPLEMENTATION** specifies the requirements for this implementation. **X12 STANDARD** is included as a reference only.

The transaction set presentation is comprised of two main sections with subsections within the main sections:

2.3 Transaction Set Listing

There are two sub-sections under this general title. The first sub-section concerns this implementation of a generic X12 transaction set. The second sub-section concerns the generic X12 standard itself.

IMPLEMENTATION

This section lists the levels, loops, and segments contained in this implementation. It also serves as an index to the segment detail.

STANDARD

This section is included as a reference.

2.4 Segment Detail

There are three sub-sections under this general title. This section repeats once for each segment used in this implementation providing segment specific detail and X12 standard detail.

SEGMENT DETAIL

This section is included as a reference.

DIAGRAM

This section is included as a reference. It provides a pictorial view of the standard and shows which elements are used in this implementation.

ELEMENT DETAIL

This section specifies the implementation details of each data element.

These illustrations (Figures 2.1 through 2.5) are examples and are not extracted from the Section 2 detail in this implementation guide. Annotated illustrations, presented below in the same order they appear in this implementation guide, describe the format of the transaction set that follows.

IMPLEMENTATION

Indicates that this section is the implementation and not the standard

8XX Insurance Transaction Set

Table 1 - Header

PAGE #	POS. #	SEG. ID	NAME	USAGE	REPEAT	LOOP REPEAT
53	0100	ST	Transaction Set Header	R	1	
54	0200	BPR	Financial Information	R	1	
60	0400	TRN	Reassociation Key	R	1	
62	0500	CUR	Non-US Dollars Currency	S	1	
65	0600	REF	Receiver ID	S	1	
66	0600	REF	Version Number	S	1	
68	0700	DTM	Production Date	S	1	
PAYER NAME						1
70	0800	N1	Payer Name	R	1	
72	1000	N3	Payer Address	S	1	
75	1100	N4	Payer City, State, Zip	S	1	
76	1200	REF	Additional Payer Reference Number	S	1	
78	1300	PER	Payer Contact	S	1	
PAYEE NAME						1
79	0800	N1	Payee Name	R	1	
81	1000	N3	Payee Address	S	1	
82	1100	N4	Payee City, State, Zip	S	1	
84	1200	REF	Payee Additional Reference Number	S	>1	

Position Numbers and Segment IDs retain their X12 values

Individual segments and entire loops are repeated

Figure 2.1. Transaction Set Key — Implementation

STANDARD						
Indicates that this section is identical to the ASC X12 standard 8XX Insurance Transaction Set Functional Group ID: XX See <i>Appendix B.1, ASC X12 Nomenclature</i> for a complete description of the standard This Draft Standard for Trial Use contains the format and establishes the data contents of the Insurance Transaction Set (8XX) within the context of the Electronic Data Interchange (EDI) environment.						
Table 1 - Header						
POS. #	SEG. ID	NAME	REQ. DES.	MAX USE	LOOP REPEAT	
0100	ST	Transaction Set Header	M	1		
0200	BPR	Beginning Segment	M	1		
0300	NTE	Note/Special Instruction	O	>1		
0400	TRN	Trace	O	1		

Figure 2.2. Transaction Set Key — Standard

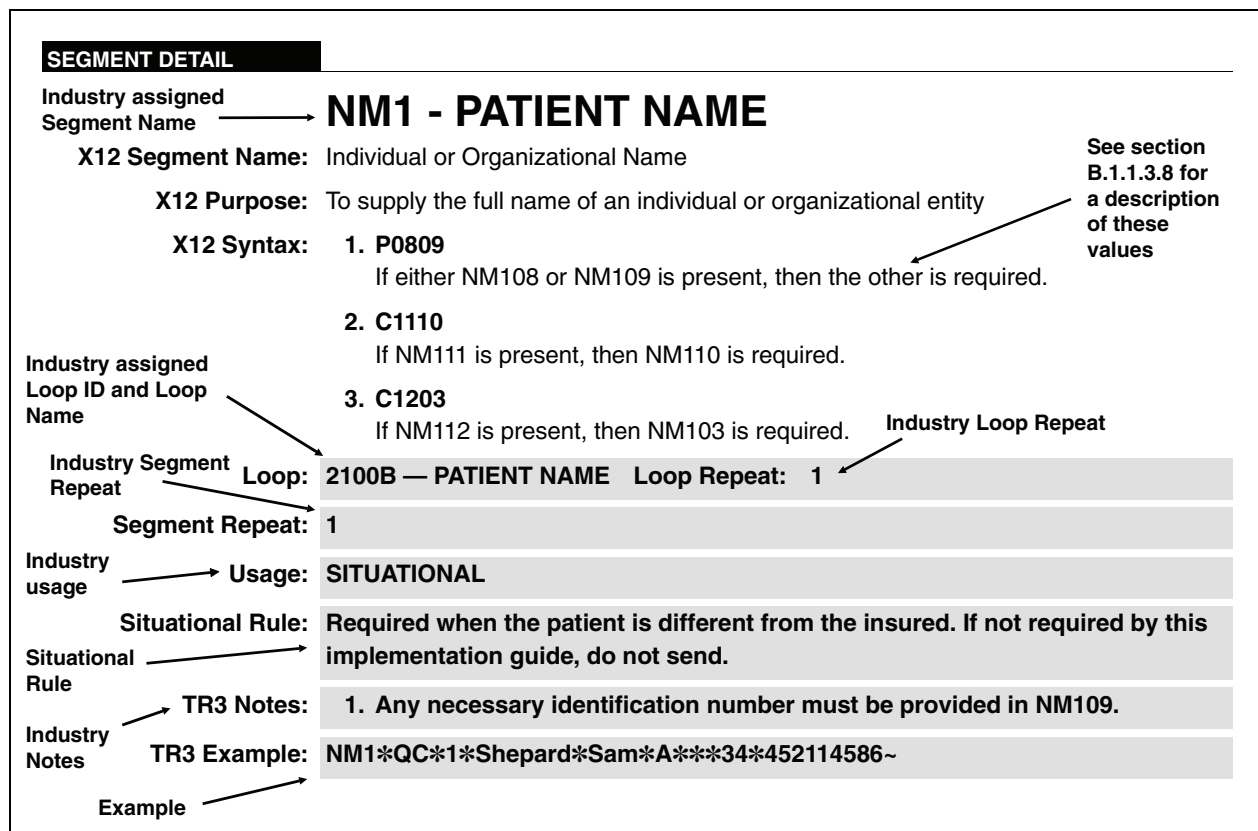


Figure 2.3. Segment Key — Implementation

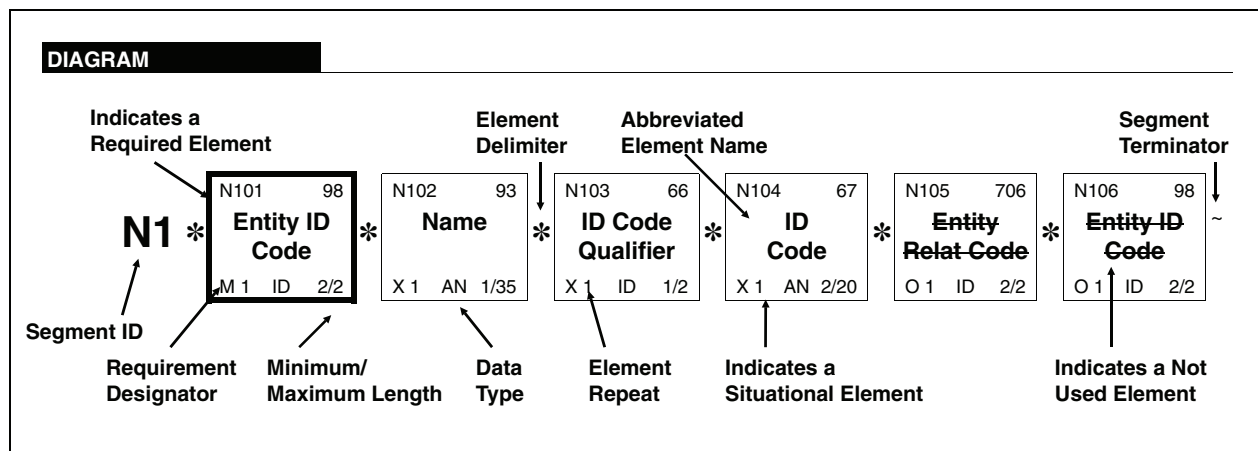


Figure 2.4. Segment Key — Diagram

ELEMENT DETAIL						
USAGE	REF. DES.	DATA ELEMENT	NAME	Element Repeat	ATTRIBUTES	
REQUIRED	SVC01	C003	COMPOSITE MEDICAL PROCEDURE IDENTIFIER To identify a medical procedure by its standardized codes and applicable modifiers Use the Primary Payer's adjudicated Medical Procedure Code.	M 1		
Reference Designator			Composite Number			
REQUIRED	SVC01 - 1	235	Product/Service ID Qualifier Code identifying the type/source of the descriptive number used in Product/Service ID (234) IMPLEMENTATION NAME: Product or Service ID Qualifier The value in SVC01-1 qualifies the values in SVC01-2, SVC01-3, SVC01-4, SVC01-5, and SVC01-6.	M ID	2/2	
Industry Usage: See the following page for complete descriptions			Industry Note			
Selected Code Values			CODE	DEFINITION		
			AD	American Dental Association Codes CODE SOURCE 135: American Dental Association		
See Appendix A for external code source reference			HP	Health Insurance Prospective Payment System (HIPPS) Skilled Nursing Facility Rate Code CODE SOURCE 716: Health Insurance Prospective Payment System (HIPPS) Rate Code for Skilled Nursing Facilities		
REQUIRED	SVC01 - 2	234	Product/Service ID Identifying number for a product or service	M AN	1/48	
NOT USED	SVC01 - 3	1339	Procedure Modifier	O AN	2/2	
NOT USED	SVC01 - 4	1339	Procedure Modifier	O AN	2/2	
NOT USED	SVC01 - 5	1339	Procedure Modifier	O AN	2/2	
NOT USED	SVC01 - 6	1339	Procedure Modifier	O AN	2/2	
NOT USED	SVC01 - 7	352	Description	O AN	1/80	
REQUIRED	SVC02	782	Monetary Amount Monetary amount SEMANTIC: SVC02 is the submitted service charge. This value can not be negative.	M 1 R	1/18	
Data Element Number						
NOT USED	SVC03	782	Monetary Amount	O 1 R	1/18	
SITUATIONAL	SVC04	234	Product/Service ID Identifying number for a product or service SEMANTIC: SVC04 is the National Uniform Billing Committee Revenue Code. SITUATIONAL RULE: Required when an NUBC revenue code was considered during adjudication in addition to a procedure code already identified in SVC01. If not required by this implementation guide, do not send. IMPLEMENTATION NAME: National Uniform Billing Committee Revenue Code	O 1 AN	1/48	
X12 Semantic Note						
Situational Rule						
Implementation Name See Appendix E for definition						

Figure 2.5. Segment Key — Element Summary

2.2 Implementation Usage

2.2.1 Industry Usage

Industry Usage describes when loops, segments, and elements are to be sent when complying with this implementation guide. The three choices for Usage are required, not used, and situational. To avoid confusion, these are named differently than the X12 standard Condition Designators (mandatory, optional, and relational).

Required This loop/segment/element must always be sent.

Required segments in Situational loops only occur when the loop is used.

Required elements in Situational segments only occur when the segment is used.

Required component elements in Situational composite elements only occur when the composite element is used.

Not Used This element must never be sent.

Situational Use of this loop/segment/element varies, depending on data content and business context as described in the defining rule. The defining rule is documented in a Situational Rule attached to the item.

There are two forms of Situational Rules.

The first form is “Required when <explicit condition statement>. If not required by this implementation guide, may be provided at the sender’s discretion, but cannot be required by the receiver.” The data qualified by such a situational rule cannot be required or requested by the receiver, transmission of this data is solely at the sender’s discretion.

The alternative form is “Required when <explicit condition statement>. If not required by this implementation guide, do not send.” The data qualified by such a situational rule cannot be sent except as described in the explicit condition statement.

2.2.1.1**Transaction Compliance Related to Industry Usage**

A transmitted transaction complies with an implementation guide when it satisfies the requirements as defined within the implementation guide. The presence or absence of an item (loop, segment, or element) complies with the industry usage specified by this implementation guide according to the following table.

Industry Usage	Business Condition is	Item is	Transaction Complies with Implementation Guide?
Required	N/A	Sent	Yes
		Not Sent	No
Not Used	N/A	Sent	No
		Not Sent	Yes
Situational (Required when <explicit condition statement>. If not required by this implementation guide, may be provided at the sender's discretion, but cannot be required by the receiver.)	True	Sent	Yes
		Not Sent	No
	Not True	Sent	Yes
		Not Sent	Yes
Situational (Required when <explicit condition statement>. If not required by this implementation guide, do not send.)	True	Sent	Yes
		Not Sent	No
	Not True	Sent	No
		Not Sent	Yes

This table specifies how an entity is to evaluate a transmitted transaction for compliance with industry usage. It is not intended to require or imply that the receiver must reject non-compliant transactions. The receiver will handle non-compliant transactions based on its business process and any applicable regulations.

2.2.2**Loops**

Loop requirements depend on the context or location of the loop within the transaction. See Appendix B for more information on loops.

- A nested loop can be used only when the associated higher level loop is used.
- The usage of a loop is the same as the usage of its beginning segment.
 - If a loop's beginning segment is Required, the loop is Required and must occur at least once unless it is nested in a loop that is not being used.
 - If a loop's beginning segment is Situational, the loop is Situational.
- Subsequent segments within a loop can be sent only when the beginning segment is used.
- Required segments in Situational loops occur only when the loop is used.

2.3 Transaction Set Listing

2.3.1 Implementation

This section lists the levels, loops, and segments contained in this implementation. It also serves as an index to the segment detail. Refer to section 2.1 Presentation Examples for detailed information on the components of the Implementation section.

IMPLEMENTATION**999** Implementation Acknowledgment**Table 1 - Header**

PAGE #	POS. #	SEG. ID	NAME	USAGE	REPEAT	LOOP REPEAT
	0100	ST	Transaction Set Header	R	1	
	0200	AK1	Functional Group Response Header	R	1	
			LOOP ID - 2000 - AK2 TRANSACTION SET RESPONSE HEADER			>1
	0300	AK2	Transaction Set Response Header	S	1	
			LOOP ID - 2100 - AK2/IK3 ERROR IDENTIFICATION			>1
	0400	IK3	Error Identification	S	1	
	0500	CTX	Segment Context	S	9	
	0500	CTX	Business Unit Identifier	S	1	
			LOOP ID - 2110 - AK2/IK3/IK4 IMPLEMENTATION DATA ELEMENT NOTE			>1
	0600	IK4	Implementation Data Element Note	S	1	
	0700	CTX	Element Context	S	10	
	0800	IK5	Transaction Set Response Trailer	R	1	
	0900	AK9	Functional Group Response Trailer	R	1	
	1000	SE	Transaction Set Trailer	R	1	

2.3.2 X12 Standard

This section is included as a reference. The implementation guide reference clarifies actual usage. Refer to section 2.1 Presentation Examples for detailed information on the components of the X12 Standard section.

STANDARD

999 Implementation Acknowledgment

Functional Group ID: **FA**

This X12 Transaction Set contains the format and establishes the data contents of the Implementation Acknowledgment Transaction Set (999) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to define the control structures for a set of acknowledgments to indicate the results of the syntactical and relational analysis of the electronically encoded documents, based upon a full or implemented subset of X12 transaction sets. The encoded documents are the transaction sets, which are grouped in functional groups, used in defining transactions for business data interchange. This standard does not cover the semantic meaning of the information encoded in the transaction sets.

Table 1 - Header

POS. #	SEG. ID	NAME	REQ. DES.	MAX USE	LOOP REPEAT
0100	ST	Transaction Set Header	M	1	
0200	AK1	Functional Group Response Header	M	1	
		LOOP ID - AK2			>1
0300	AK2	Transaction Set Response Header	O	1	
		LOOP ID - AK2/IK3			>1
0400	IK3	Implementation Data Segment Note	O	1	
0500	CTX	Context	O	10	
		LOOP ID - AK2/IK3/IK4			>1
0600	IK4	Implementation Data Element Note	O	1	
0700	CTX	Context	O	10	
0800	IK5	Implementation Transaction Set Response Trailer	M	1	
0900	AK9	Functional Group Response Trailer	M	1	
1000	SE	Transaction Set Trailer	M	1	

NOTES:

- 1/0100** Neither the 997 nor the 999 Acknowledgment shall be acknowledged, thereby preventing an endless cycle of acknowledgments of acknowledgments. Nor shall a Implementation Acknowledgment be sent to report errors in a previous Implementation Acknowledgment.
- 1/0100** There is only one Implementation Acknowledgment Transaction Set per acknowledged functional group.
- 1/0100** Only one acknowledgement, either a single Transaction Set 997 or a single Transaction Set 999, should be generated for a functional group unless mutually agreed upon.
- 1/0200** AK1 is used to respond to the functional group header and to start the acknowledgment for a functional group. There shall be one AK1 segment for the functional group that is being acknowledged.
- 1/0200** The Implementation Acknowledgement is generated at the point of translation, intended for the originator (not any intermediate parties).
- 1/0200** The Functional Group Header Segment (GS) is used to start the envelope for the Implementation Acknowledgment Transaction Sets. In preparing the functional group of acknowledgments, the application sender's code and the application receiver's code, taken from the functional group being acknowledged, are exchanged; therefore, one acknowledgment functional group responds to only those functional groups from one application receiver's code to one application sender's code.
- 1/0300** AK2 is used to start the acknowledgment of a transaction set within the received functional group. The AK2 segments shall appear in the same order as the transaction sets in the functional group that has been received and is being acknowledged.

- 1/0400** The data segments of this standard are used to report the results of the syntactical analysis of the functional groups of transaction sets; they report the extent to which the syntax complies with the standards or proper subsets of transaction sets and functional groups as expressed in compliant implementation guides. They do not report on the semantic meaning of the transaction sets (for example, on the ability of the receiver to comply with the request of the sender).
- 1/0700** The CTX Segment shall be used to disambiguate a reported error that is dependent on context.
- 1/0800** If any implementation guide errors have been reported in IK3 or IK4, then code I5 shall be reported in the IK5 Segment.

2.4 Segment Detail

This section specifies the segments, data elements, and codes for this implementation. Refer to section 2.1 Presentation Examples for detailed information on the components of the Segment Detail section.

SEGMENT DETAIL

ST - TRANSACTION SET HEADER**X12 Segment Name:** Transaction Set Header**X12 Purpose:** To indicate the start of a transaction set and to assign a control number

- X12 Set Notes:**
1. Neither the 997 nor the 999 Acknowledgment shall be acknowledged, thereby preventing an endless cycle of acknowledgments of acknowledgments. Nor shall a Implementation Acknowledgment be sent to report errors in a previous Implementation Acknowledgment.
 2. There is only one Implementation Acknowledgment Transaction Set per acknowledged functional group.
 3. Only one acknowledgement, either a single Transaction Set 997 or a single Transaction Set 999, should be generated for a functional group unless mutually agreed upon.

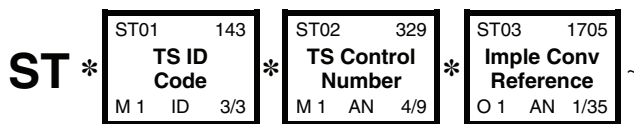
Segment Repeat: 1**Usage:** REQUIRED

TR3 Notes:

1. When acknowledging a healthcare implementation guide (TR3) based transaction, only a single transaction set 999 can be used. When acknowledging a healthcare implementation guide (TR3) based transaction, the transaction set 997 must not be used.

TR3 Example: ST*999*0001*005010X231A1~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	ST01	143	Transaction Set Identifier Code Code uniquely identifying a Transaction Set SEMANTIC: The transaction set identifier (ST01) is used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).	M 1 ID 3/3
			CODE	DEFINITION
			999	Implementation Acknowledgment
REQUIRED	ST02	329	Transaction Set Control Number Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set The Transaction Set Control Numbers in ST02 and SE02 must be identical. The number is assigned by the originator and must be unique within a functional group (GS-GE). The number also aids in error resolution research. For example, start with the number 0001 and increment from there.	M 1 AN 4/9

REQUIRED	ST03	1705	Implementation Convention Reference Reference assigned to identify Implementation Convention	O 1	AN	1/35
----------	------	------	--	-----	----	------

SEMANTIC: The implementation convention reference (ST03) is used by the translation routines of the interchange partners to select the appropriate implementation convention to match the transaction set definition. When used, this implementation convention reference takes precedence over the implementation reference specified in the GS08.

This field contains the same value as data element GS08. This value is always 005010X231A1 when this implementation guide is utilized. Some translator products strip off the ISA and GS segments prior to application processing. Providing the information from GS08 at this level will help ensure the appropriate application mapping is utilized at translation time.

SEGMENT DETAIL

AK1 - FUNCTIONAL GROUP RESPONSE HEADER

X12 Segment Name: Functional Group Response Header

X12 Purpose: To start acknowledgment of a functional group

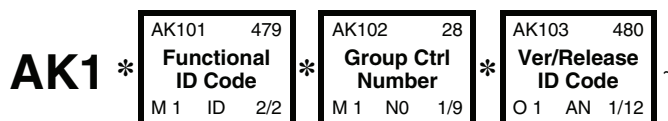
- X12 Set Notes:**
1. AK1 is used to respond to the functional group header and to start the acknowledgment for a functional group. There shall be one AK1 segment for the functional group that is being acknowledged.
 2. The Implementation Acknowledgement is generated at the point of translation, intended for the originator (not any intermediate parties).
 3. The Functional Group Header Segment (GS) is used to start the envelope for the Implementation Acknowledgment Transaction Sets. In preparing the functional group of acknowledgments, the application sender's code and the application receiver's code, taken from the functional group being acknowledged, are exchanged; therefore, one acknowledgment functional group responds to only those functional groups from one application receiver's code to one application sender's code.

Segment Repeat: 1

Usage: REQUIRED

TR3 Example: AK1*HC*0001*004010X098A1~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	AK101	479	Functional Identifier Code Code identifying a group of application related transaction sets SEMANTIC: AK101 is the functional ID found in the GS segment (GS01) in the functional group being acknowledged. Use the value in GS01 from the functional group to which this 999 transaction set is responding.	M 1 ID 2/2
REQUIRED	AK102	28	Group Control Number Assigned number originated and maintained by the sender SEMANTIC: AK102 is the functional group control number found in the GS segment in the functional group being acknowledged. Use the value in GS06 from the functional group to which this 999 transaction set is responding.	M 1 N0 1/9

REQUIRED	AK103	480	Version / Release / Industry Identifier Code Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed SEMANTIC: AK103 is the version release industry identifier code in the GS segment (GS08) in the functional group being acknowledged. IMPLEMENTATION NAME: Version, Release, or Industry Identifier Code CODE SOURCE 881: Version / Release / Industry Identifier Code Use the value in GS08 from the functional group to which this 999 transaction set is responding.	O 1 AN 1/12
-----------------	--------------	------------	--	--------------------

SEGMENT DETAIL

AK2 - TRANSACTION SET RESPONSE HEADER

X12 Segment Name: Transaction Set Response Header

X12 Purpose: To start acknowledgment of a single transaction set

X12 Set Notes: 1. AK2 is used to start the acknowledgment of a transaction set within the received functional group. The AK2 segments shall appear in the same order as the transaction sets in the functional group that has been received and is being acknowledged.

Loop: 2000 - AK2 — TRANSACTION SET RESPONSE HEADER **Loop Repeat:** >1

Segment Repeat: 1

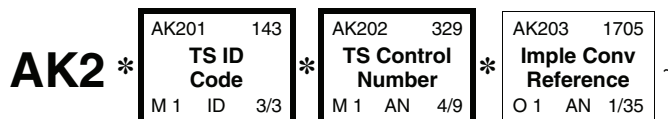
Usage: SITUATIONAL

Situational Rule: Required when an error is present in a transaction set contained in the functional group to which this 999 transaction set is responding. If not required by this implementation guide, may be provided at the sender's discretion but cannot be required by the receiver.

TR3 Notes: 1. While it is not required that an AK2 loop be included for each transaction set received, it is required that an AK2 loop be included for each transaction set that contains an error. When AK2 loops are included in this transaction set, they must be in the same order as received in the functional group to which this 999 is responding.

TR3 Example: AK2*837*0001~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	AK201	143	Transaction Set Identifier Code Code uniquely identifying a Transaction Set	M 1 ID 3/3
SEMANTIC: AK201 is the transaction set ID found in the ST segment (ST01) in the transaction set being acknowledged.				
Use the value in ST01 from the transaction set to which this 999 transaction set is responding.				

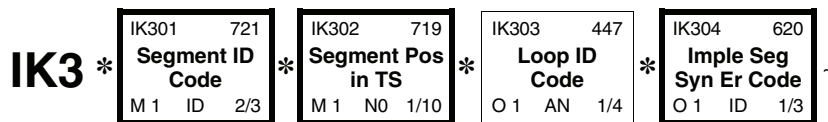
REQUIRED	AK202	329	Transaction Set Control Number M 1 AN 4/9 Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set SEMANTIC: AK202 is the transaction set control number found in the ST segment in the transaction set being acknowledged. Use the value in ST02 from the transaction set to which this 999 transaction set is responding.
SITUATIONAL	AK203	1705	Implementation Convention Reference O 1 AN 1/35 Reference assigned to identify Implementation Convention SEMANTIC: AK203 is the implementation convention reference, if any, found in the ST segment (ST03) in the transaction set being acknowledged. SITUATIONAL RULE: <i>Required when the ST03 value is available in the transaction set to which this 999 transaction set is responding. If not required by this implementation guide, do not send.</i> When used, this is the value in ST03 from the transaction set to which this 999 transaction set is responding.

SEGMENT DETAIL

IK3 - ERROR IDENTIFICATION

X12 Segment Name: Implementation Data Segment Note**X12 Purpose:** To report implementation errors in a data segment and identify the location of the data segment**X12 Set Notes:** 1. The data segments of this standard are used to report the results of the syntactical analysis of the functional groups of transaction sets; they report the extent to which the syntax complies with the standards or proper subsets of transaction sets and functional groups as expressed in compliant implementation guides. They do not report on the semantic meaning of the transaction sets (for example, on the ability of the receiver to comply with the request of the sender).**Loop:** 2100 - AK2/IK3 — ERROR IDENTIFICATION **Loop Repeat:** >1**Segment Repeat:** 1**Usage:** SITUATIONAL**Situational Rule:** Required when an error is present in the transaction set identified in this AK2 loop and the location of the data segment containing the error can be identified by the submitter of this 999. If not required by this implementation guideline, do not send.**TR3 Example:** IK3*DMG*31*2000*8~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	IK301	721	Segment ID Code Code defining the segment ID of the data segment in error (See Appendix A - Number 77)	M 1 ID 2/3
			CODE SOURCE 77: X12 Directories	
REQUIRED	IK302	719	Segment Position in Transaction Set The numerical count position of this data segment from the start of the transaction set: the transaction set header is count position 1	M 1 NO 1/10
			The value to use in IK302 is the numerical count position of this data segment, relative to the transaction set instance (not the transaction set diagram), from the start of the transaction set. The transaction set header (i.e. ST) is count position 1.	

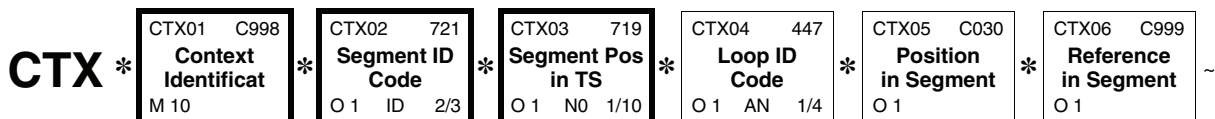
SITUATIONAL	IK303	447	Loop Identifier Code	O 1	AN	1/4
The loop ID number given on the transaction set diagram is the value for this data element in segments LS and LE						
SITUATIONAL RULE: <i>Required when the data segment containing the error is within an LS loop and the LS01 value is known by the submitter of this 999, or when the data segment containing the error is within another type of loop and the loop identifier value from the transaction set diagram is known by the submitter of this 999. If not required by this implementation guide, do not send.</i>						
Only the first four characters of the loop identifier must be used.						
REQUIRED	IK304	620	Implementation Segment Syntax Error Code	O 1	ID	1/3
Code indicating implementation error found based on the syntax editing of a segment						
			CODE	DEFINITION		
			1	Unrecognized segment ID		
			2	Unexpected segment		
			3	Required Segment Missing		
			4	Loop Occurs Over Maximum Times		
			5	Segment Exceeds Maximum Use		
			6	Segment Not in Defined Transaction Set		
			7	Segment Not in Proper Sequence		
			8	Segment Has Data Element Errors		
			I4	Implementation “Not Used” Segment Present		
			I6	Implementation Dependent Segment Missing		
			I7	Implementation Loop Occurs Under Minimum Times		
			I8	Implementation Segment Below Minimum Use		
			I9	Implementation Dependent “Not Used” Segment Present		

SEGMENT DETAIL

CTX - SEGMENT CONTEXT

X12 Segment Name: Context**X12 Purpose:** Describes an event context in terms of the application or implementation contexts in force at the time the event occurred and the position in the EDI stream at which that context was activated**Loop:** 2100 - AK2/IK3 — ERROR IDENTIFICATION**Segment Repeat:** 9**Usage:** SITUATIONAL**Situational Rule:** Required when the error identified in this IK3 loop was triggered by a situational requirement of the implementation guide and the error occurs at the segment level. If not required by this implementation guide, do not send.**TR3 Notes:** 1. The CTX segment is used to identify the data that triggered the situational requirement.**TR3 Example:** CTX*SITUATIONAL TRIGGER*CLM*43**5:3~**TR3 Example:** CTX*SITUATIONAL TRIGGER*CLM*43**2*782~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	CTX01	C998	CONTEXT IDENTIFICATION	M 10
			Holds information to identify a context	
REQUIRED	CTX01 - 1	9999	Context Name	M AN 1/35
			Holds the name or 'tag' of a context	
			Always contains the value "SITUATIONAL TRIGGER".	
NOT USED	CTX01 - 2	9998	Context Reference	O AN 1/35
REQUIRED	CTX02	721	Segment ID Code	O 1 ID 2/3
			Code defining the segment ID of the data segment in error (See Appendix A - Number 77)	
			CODE SOURCE 77: X12 Directories	
REQUIRED	CTX03	719	Segment Position in Transaction Set	O 1 NO 1/10
			The numerical count position of this data segment from the start of the transaction set: the transaction set header is count position 1	

SITUATIONAL	CTX04	447	Loop Identifier Code	O 1 AN 1/4
The loop ID number given on the transaction set diagram is the value for this data element in segments LS and LE				
SITUATIONAL RULE: <i>Required when the data segment containing the situational requirement is within an LS loop and the LS01 value is known by the submitter of this 999, or when the data segment containing the situational requirement is within another type of loop and the loop identifier value from the transaction set diagram is known by the submitter of this 999. If not required by this implementation guide, do not send.</i>				
Only the first four characters of the loop identifier must be used.				
SITUATIONAL	CTX05	C030	POSITION IN SEGMENT	O 1
Code indicating the relative position of the simple data element or composite data structure in error within a segment, count beginning with 1 for the position immediately following the segment ID; additionally indicating the relative position of a repeating structure in error, count beginning with 1 for the position immediately following the preceding element separator; additionally indicating the relative position of a component of a composite data structure in error, count beginning with 1 for the position following the preceding element or repetition separator				
SITUATIONAL RULE: <i>Required when the situational requirement relates to an element. If not required by this implementation guide, do not send.</i>				
REQUIRED	CTX05 - 1	722	Element Position in Segment	M N0 1/2
This is used to indicate the relative position of a simple data element, or the relative position of a composite data structure with the relative position of the component within the composite data structure, in error; in the data segment the count starts with 1 for the simple data element or composite data structure immediately following the segment ID				
SITUATIONAL	CTX05 - 2	1528	Component Data Element Position in Composite	O N0 1/2
To identify the component data element position within the composite that is in error				
SITUATIONAL RULE: <i>Required when the situational requirement relates to a component data element within a composite data structure. If not required by this implementation guide, do not send.</i>				
SITUATIONAL	CTX05 - 3	1686	Repeating Data Element Position	O N0 1/4
To identify the specific repetition of a data element that is in error				
SITUATIONAL RULE: <i>Required when the situational requirement relates to a repeating data element. If not required by this implementation guide, do not send.</i>				
SITUATIONAL	CTX06	C999	REFERENCE IN SEGMENT	O 1
To hold the reference number of a data element and optionally a component data element within a composite				
SITUATIONAL RULE: <i>Required when CTX05 is used and the data element reference number of the data element identified in CTX05-1 is known by the submitter of the 999, and it is NOT a composite data element. If not required by this implementation guide, do not send.</i>				

REQUIRED	CTX06 - 1	725	Data Element Reference Number	M	N0	1/4
			Reference number used to locate the data element in the Data Element Dictionary			
			SEMANTIC: This element holds the reference number of the simple or composite element at segment level.			
			CODE SOURCE 77 : X12 Directories			
NOT USED	CTX06 - 2	725	Data Element Reference Number	O	N0	1/4

SEGMENT DETAIL

CTX - BUSINESS UNIT IDENTIFIER

X12 Segment Name: Context**X12 Purpose:** Describes an event context in terms of the application or implementation contexts in force at the time the event occurred and the position in the EDI stream at which that context was activated**Loop:** 2100 - AK2/IK3 — ERROR IDENTIFICATION**Segment Repeat:** 1**Usage:** SITUATIONAL**Situational Rule:** Required when the error reported in this IK3 loop is within a business unit and the business unit identifier is known by the submitter of the 999. If not required by this implementation guide, do not send.

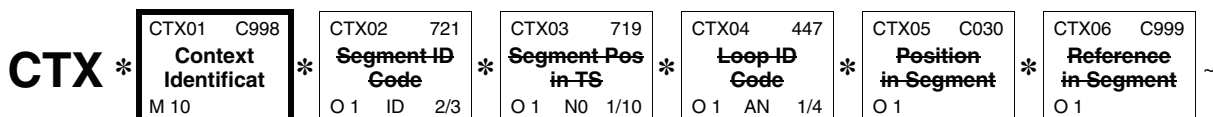
TR3 Notes:

1. Valid values for the business unit identifier are:

TRN02	269 business unit identifier
TRN02	270 business unit identifier
TRN02	271 business unit identifier
NM109	274 business unit identifier
PATIENT NAME NM109	275 business unit identifier
TRN02	276 business unit identifier
TRN02	277 business unit identifier
SUBSCRIBER NAME NM109	278 business unit identifier
ENT01	820 business unit identifier
SUBSCRIBER NUMBER REF02	834 business unit identifier
TRN02	835 business unit identifier
CLM01	837 business unit identifier

TR3 Example: CTX*CLM01:123456789~

DIAGRAM



ELEMENT DETAIL

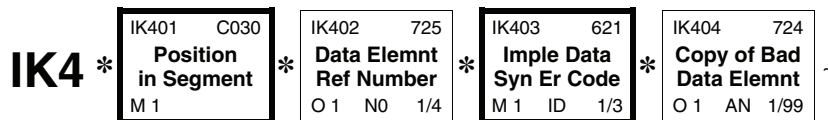
USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	CTX01	C998	CONTEXT IDENTIFICATION	M 10
Holds information to identify a context				

REQUIRED	CTX01 - 1	9999	Context Name Holds the name or 'tag' of a context	M	AN	1/35
This contains one of the following values, depending on the value in AK201: TRN02 269 business unit identifier TRN02 270 business unit identifier TRN02 271 business unit identifier NM109 274 business unit identifier PATIENT NAME NM109 275 business unit identifier TRN02 276 business unit identifier TRN02 277 business unit identifier SUBSCRIBER NAME NM109 278 business unit identifier ENT01 820 business unit identifier SUBSCRIBER NUMBER REF02 834 business unit identifier TRN02 835 business unit identifier CLM01 837 business unit identifier						
REQUIRED	CTX01 - 2	9998	Context Reference Holds a reference to or for a context	O	AN	1/35
This contains the value from the business unit identifier specified in CTX01-1.						
NOT USED	CTX02	721	Segment ID Code	O 1	ID	2/3
NOT USED	CTX03	719	Segment Position in Transaction Set	O 1	N0	1/10
NOT USED	CTX04	447	Loop Identifier Code	O 1	AN	1/4
NOT USED	CTX05	C030	POSITION IN SEGMENT	O 1		
NOT USED	CTX06	C999	REFERENCE IN SEGMENT	O 1		

SEGMENT DETAIL

IK4 - IMPLEMENTATION DATA ELEMENT
NOTE**X12 Segment Name:** Implementation Data Element Note**X12 Purpose:** To report implementation errors in a data element or composite data structure and identify the location of the data element**Loop:** 2110 - AK2/IK3/IK4 — IMPLEMENTATION DATA ELEMENT NOTE **Loop Repeat:** >1**Segment Repeat:** 1**Usage:** SITUATIONAL**Situational Rule:** Required when the error in the segment described in the IK3 segment applies to a data element and the location of the data element containing the error can be identified by the submitter of the 999. If not required by this implementation guideline, do not send.**TR3 Example:** IK4*3*1068*7*B~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	IK401	C030	POSITION IN SEGMENT	M 1 Code indicating the relative position of the simple data element or composite data structure in error within a segment, count beginning with 1 for the position immediately following the segment ID; additionally indicating the relative position of a repeating structure in error, count beginning with 1 for the position immediately following the preceding element separator; additionally indicating the relative position of a component of a composite data structure in error, count beginning with 1 for the position following the preceding element or repetition separator
REQUIRED	IK401 - 1	722	Element Position in Segment	M N0 1/2 This is used to indicate the relative position of a simple data element, or the relative position of a composite data structure with the relative position of the component within the composite data structure, in error; in the data segment the count starts with 1 for the simple data element or composite data structure immediately following the segment ID
SITUATIONAL	IK401 - 2	1528	Component Data Element Position in Composite	O N0 1/2 To identify the component data element position within the composite that is in error SITUATIONAL RULE: <i>Required when the error described in this segment relates to a component data element within a composite data structure. If not required by this implementation guide, do not send.</i>

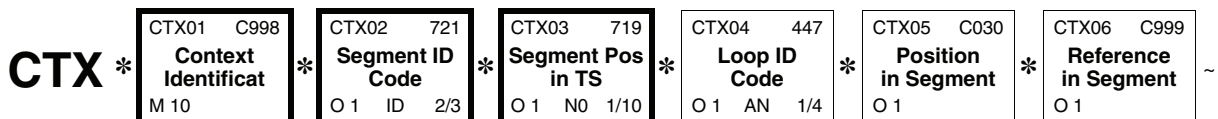
SITUATIONAL	IK401 - 3	1686	Repeating Data Element Position		O	N0	1/4
			To identify the specific repetition of a data element that is in error				
			SITUATIONAL RULE: <i>Required when the error described in this segment relates to a repeating data element. If not required by this implementation guide, do not send.</i>				
SITUATIONAL	IK402	725	Data Element Reference Number		O	1	N0 1/4
			Reference number used to locate the data element in the Data Element Dictionary				
			SITUATIONAL RULE: <i>Required when the data element reference number for the errored data is known. If not required by this implementation guide, do not send.</i>				
			CODE SOURCE 77: X12 Directories				
REQUIRED	IK403	621	Implementation Data Element Syntax Error Code		M	1	ID 1/3
			Code indicating the implementation error found after syntax edits of a data element				
			CODE	DEFINITION			
			1	Required Data Element Missing			
			2	Conditional Required Data Element Missing			
			3	Too Many Data Elements			
			4	Data Element Too Short			
			5	Data Element Too Long			
			6	Invalid Character In Data Element			
			7	Invalid Code Value			
			8	Invalid Date			
			9	Invalid Time			
			10	Exclusion Condition Violated			
			12	Too Many Repetitions			
			13	Too Many Components			
			I10	Implementation “Not Used” Data Element Present			
			I11	Implementation Too Few Repetitions			
			I12	Implementation Pattern Match Failure			
			I13	Implementation Dependent “Not Used” Data Element Present			
			I6	Code Value Not Used in Implementation			
			I9	Implementation Dependent Data Element Missing			
SITUATIONAL	IK404	724	Copy of Bad Data Element		O	1	AN 1/99
			This is a copy of the data element in error				
			SITUATIONAL RULE: <i>Required unless invalid characters are present, data is missing, or the data is from a binary data element. If not required by this implementation guide, do not send.</i>				

SEGMENT DETAIL

CTX - ELEMENT CONTEXT

X12 Segment Name: Context**X12 Purpose:** Describes an event context in terms of the application or implementation contexts in force at the time the event occurred and the position in the EDI stream at which that context was activated**X12 Set Notes:** 1. The CTX Segment shall be used to disambiguate a reported error that is dependent on context.**Loop:** 2110 - AK2/IK3/IK4 — IMPLEMENTATION DATA ELEMENT NOTE**Segment Repeat:** 10**Usage:** SITUATIONAL**Situational Rule:** Required when the error identified in this IK4 loop was triggered by a situational requirement of the implementation guide and the error occurs at the element level. If not required by this implementation guide, do not send.**TR3 Notes:** 1. The CTX segment is used to identify the data that triggered the situational requirement.**TR3 Example:** CTX*SITUATIONAL TRIGGER*CLM*43**5:3~**TR3 Example:** CTX*SITUATIONAL TRIGGER*CLM*43**2*782~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	CTX01	C998	CONTEXT IDENTIFICATION	M 10
			Holds information to identify a context	
REQUIRED	CTX01 - 1	9999	Context Name	M AN 1/35
			Holds the name or 'tag' of a context	
			Always contains the value "SITUATIONAL TRIGGER".	
NOT USED	CTX01 - 2	9998	Context Reference	O AN 1/35
REQUIRED	CTX02	721	Segment ID Code	O 1 ID 2/3
			Code defining the segment ID of the data segment in error (See Appendix A - Number 77)	
			CODE SOURCE 77: X12 Directories	
REQUIRED	CTX03	719	Segment Position in Transaction Set	O 1 NO 1/10
			The numerical count position of this data segment from the start of the transaction set: the transaction set header is count position 1	

SITUATIONAL	CTX04	447	Loop Identifier Code	O 1 AN 1/4
The loop ID number given on the transaction set diagram is the value for this data element in segments LS and LE				
SITUATIONAL RULE: <i>Required when the data segment containing the situational requirement is within an LS loop and the LS01 value is known by the submitter of this 999, or when the data segment containing the situational requirement is within another type of loop and the loop identifier value from the transaction set diagram is known by the submitter of this 999. If not required by this implementation guide, do not send.</i>				
Only the first four characters of the loop identifier must be used.				
SITUATIONAL	CTX05	C030	POSITION IN SEGMENT	O 1
Code indicating the relative position of the simple data element or composite data structure in error within a segment, count beginning with 1 for the position immediately following the segment ID; additionally indicating the relative position of a repeating structure in error, count beginning with 1 for the position immediately following the preceding element separator; additionally indicating the relative position of a component of a composite data structure in error, count beginning with 1 for the position following the preceding element or repetition separator				
SITUATIONAL RULE: <i>Required when the situational requirement relates to an element. If not required by this implementation guide, do not send.</i>				
REQUIRED	CTX05 - 1	722	Element Position in Segment	M N0 1/2
This is used to indicate the relative position of a simple data element, or the relative position of a composite data structure with the relative position of the component within the composite data structure, in error; in the data segment the count starts with 1 for the simple data element or composite data structure immediately following the segment ID				
SITUATIONAL	CTX05 - 2	1528	Component Data Element Position in Composite	O N0 1/2
To identify the component data element position within the composite that is in error				
SITUATIONAL RULE: <i>Required when the situational requirement relates to a component data element within a composite data structure. If not required by this implementation guide, do not send.</i>				
SITUATIONAL	CTX05 - 3	1686	Repeating Data Element Position	O N0 1/4
To identify the specific repetition of a data element that is in error				
SITUATIONAL RULE: <i>Required when the situational requirement relates to a repeating data element. If not required by this implementation guide, do not send.</i>				
SITUATIONAL	CTX06	C999	REFERENCE IN SEGMENT	O 1
To hold the reference number of a data element and optionally a component data element within a composite				
SITUATIONAL RULE: <i>Required when CTX05 is used and the data element reference number of the data element identified in CTX05-1 is known by the submitter of the 999, and it is NOT a composite data element. If not required by this implementation guide, do not send.</i>				

REQUIRED	CTX06 - 1	725	Data Element Reference Number	M	N0	1/4
			Reference number used to locate the data element in the Data Element Dictionary			
			SEMANTIC: This element holds the reference number of the simple or composite element at segment level.			
			CODE SOURCE 77: X12 Directories			
NOT USED	CTX06 - 2	725	Data Element Reference Number	O	N0	1/4

SEGMENT DETAIL

IK5 - TRANSACTION SET RESPONSE TRAILER

X12 Segment Name: Implementation Transaction Set Response Trailer

X12 Purpose: To acknowledge acceptance or rejection and report implementation errors in a transaction set

X12 Set Notes: 1. If any implementation guide errors have been reported in IK3 or IK4, then code I5 shall be reported in the IK5 Segment.

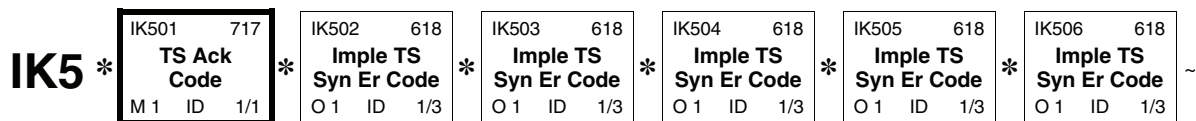
Loop: 2110 - AK2/IK3/IK4 — IMPLEMENTATION DATA ELEMENT NOTE

Segment Repeat: 1

Usage: REQUIRED

TR3 Example: IK5*R*5~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	IK501	717	Transaction Set Acknowledgment Code Code indicating accept or reject condition based on the syntax editing of the transaction set	M 1 ID 1/1
			CODE	DEFINITION
			A	Accepted
			E	Accepted But Errors Were Noted
				The transaction set indicated in this AK2 loop contained errors, but was forwarded for further processing.
			M	Rejected, Message Authentication Code (MAC) Failed
			R	Rejected
				The transaction set indicated in this AK2 loop contained errors, and was NOT forwarded for further processing. It will need to be corrected and resubmitted.
			W	Rejected, Assurance Failed Validity Tests
			X	Rejected, Content After Decryption Could Not Be Analyzed

SITUATIONAL	IK502	618	Implementation Transaction Set Syntax Error Code	O 1	ID	1/3																																																				
Code indicating implementation error found based on the syntax editing of a transaction set																																																										
SITUATIONAL RULE: <i>Required when IK501 = E or R. If not required by this implementation guide, do not send.</i>																																																										
<table><tr><th>CODE</th><th>DEFINITION</th></tr><tr><td>1</td><td>Transaction Set Not Supported</td></tr><tr><td>2</td><td>Transaction Set Trailer Missing</td></tr><tr><td>3</td><td>Transaction Set Control Number in Header and Trailer Do Not Match</td></tr><tr><td>4</td><td>Number of Included Segments Does Not Match Actual Count</td></tr><tr><td>5</td><td>One or More Segments in Error</td></tr><tr><td>6</td><td>Missing or Invalid Transaction Set Identifier</td></tr><tr><td>7</td><td>Missing or Invalid Transaction Set Control Number</td></tr><tr><td>8</td><td>Authentication Key Name Unknown</td></tr><tr><td>9</td><td>Encryption Key Name Unknown</td></tr><tr><td>10</td><td>Requested Service (Authentication or Encrypted) Not Available</td></tr><tr><td>11</td><td>Unknown Security Recipient</td></tr><tr><td>12</td><td>Incorrect Message Length (Encryption Only)</td></tr><tr><td>13</td><td>Message Authentication Code Failed</td></tr><tr><td>15</td><td>Unknown Security Originator</td></tr><tr><td>16</td><td>Syntax Error in Decrypted Text</td></tr><tr><td>17</td><td>Security Not Supported</td></tr><tr><td>18</td><td>Transaction Set not in Functional Group</td></tr><tr><td>19</td><td>Invalid Transaction Set Implementation Convention Reference</td></tr><tr><td>23</td><td>Transaction Set Control Number Not Unique within the Functional Group</td></tr><tr><td>24</td><td>S3E Security End Segment Missing for S3S Security Start Segment</td></tr><tr><td>25</td><td>S3S Security Start Segment Missing for S3E Security End Segment</td></tr><tr><td>26</td><td>S4E Security End Segment Missing for S4S Security Start Segment</td></tr><tr><td>27</td><td>S4S Security Start Segment Missing for S4E Security End Segment</td></tr><tr><td>I5</td><td>Implementation One or More Segments in Error</td></tr><tr><td>I6</td><td>Implementation Convention Not Supported</td></tr></table>							CODE	DEFINITION	1	Transaction Set Not Supported	2	Transaction Set Trailer Missing	3	Transaction Set Control Number in Header and Trailer Do Not Match	4	Number of Included Segments Does Not Match Actual Count	5	One or More Segments in Error	6	Missing or Invalid Transaction Set Identifier	7	Missing or Invalid Transaction Set Control Number	8	Authentication Key Name Unknown	9	Encryption Key Name Unknown	10	Requested Service (Authentication or Encrypted) Not Available	11	Unknown Security Recipient	12	Incorrect Message Length (Encryption Only)	13	Message Authentication Code Failed	15	Unknown Security Originator	16	Syntax Error in Decrypted Text	17	Security Not Supported	18	Transaction Set not in Functional Group	19	Invalid Transaction Set Implementation Convention Reference	23	Transaction Set Control Number Not Unique within the Functional Group	24	S3E Security End Segment Missing for S3S Security Start Segment	25	S3S Security Start Segment Missing for S3E Security End Segment	26	S4E Security End Segment Missing for S4S Security Start Segment	27	S4S Security Start Segment Missing for S4E Security End Segment	I5	Implementation One or More Segments in Error	I6	Implementation Convention Not Supported
CODE	DEFINITION																																																									
1	Transaction Set Not Supported																																																									
2	Transaction Set Trailer Missing																																																									
3	Transaction Set Control Number in Header and Trailer Do Not Match																																																									
4	Number of Included Segments Does Not Match Actual Count																																																									
5	One or More Segments in Error																																																									
6	Missing or Invalid Transaction Set Identifier																																																									
7	Missing or Invalid Transaction Set Control Number																																																									
8	Authentication Key Name Unknown																																																									
9	Encryption Key Name Unknown																																																									
10	Requested Service (Authentication or Encrypted) Not Available																																																									
11	Unknown Security Recipient																																																									
12	Incorrect Message Length (Encryption Only)																																																									
13	Message Authentication Code Failed																																																									
15	Unknown Security Originator																																																									
16	Syntax Error in Decrypted Text																																																									
17	Security Not Supported																																																									
18	Transaction Set not in Functional Group																																																									
19	Invalid Transaction Set Implementation Convention Reference																																																									
23	Transaction Set Control Number Not Unique within the Functional Group																																																									
24	S3E Security End Segment Missing for S3S Security Start Segment																																																									
25	S3S Security Start Segment Missing for S3E Security End Segment																																																									
26	S4E Security End Segment Missing for S4S Security Start Segment																																																									
27	S4S Security Start Segment Missing for S4E Security End Segment																																																									
I5	Implementation One or More Segments in Error																																																									
I6	Implementation Convention Not Supported																																																									
SITUATIONAL	IK503	618	Implementation Transaction Set Syntax Error Code	O 1	ID	1/3																																																				
Code indicating implementation error found based on the syntax editing of a transaction set																																																										
SITUATIONAL RULE: <i>Required when IK501 = E or R, and IK502 has been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>																																																										

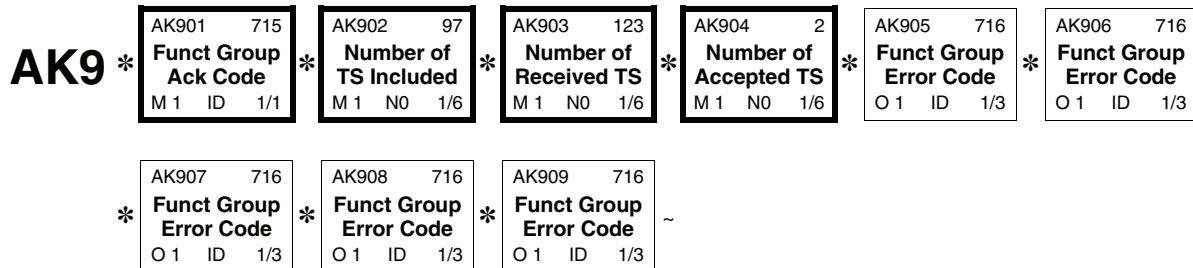
SITUATIONAL	IK504	618	Implementation Transaction Set Syntax Error Code Code indicating implementation error found based on the syntax editing of a transaction set SITUATIONAL RULE: <i>Required when IK501 = E or R, and IK502 and IK503 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3
SITUATIONAL	IK505	618	Implementation Transaction Set Syntax Error Code Code indicating implementation error found based on the syntax editing of a transaction set SITUATIONAL RULE: <i>Required when IK501 = E or R, and IK502, IK503, and IK504 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3
SITUATIONAL	IK506	618	Implementation Transaction Set Syntax Error Code Code indicating implementation error found based on the syntax editing of a transaction set SITUATIONAL RULE: <i>Required when IK501 = E or R, and IK502, IK503, IK504, and IK505 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3

SEGMENT DETAIL

AK9 - FUNCTIONAL GROUP RESPONSE TRAILER

X12 Segment Name: Functional Group Response Trailer**X12 Purpose:** To acknowledge acceptance or rejection of a functional group and report the number of included transaction sets from the original trailer, the accepted sets, and the received sets in this functional group**Segment Repeat:** 1**Usage:** REQUIRED**TR3 Example:** AK9*R*1*1*0~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES												
REQUIRED	AK901	715	Functional Group Acknowledge Code Code indicating accept or reject condition based on the syntax editing of the functional group COMMENT: If AK901 contains the value "A" or "E", then the transmitted functional group is accepted.	M 1 ID 1/1												
			<table><tr><th>CODE</th><th>DEFINITION</th></tr><tr><td>A</td><td>Accepted This code value can only be used if there are no AK2 loops or all IK501 values = 'A'.</td></tr><tr><td>E</td><td>Accepted, But Errors Were Noted. The functional group indicated in this 999 contained errors, but was forwarded for further processing.</td></tr><tr><td>M</td><td>Rejected, Message Authentication Code (MAC) Failed</td></tr><tr><td>P</td><td>Partially Accepted, At Least One Transaction Set Was Rejected</td></tr><tr><td>R</td><td>Rejected The functional group indicated in this 999 contained errors, and was NOT forwarded for further processing. It will need to be corrected and resubmitted.</td></tr></table>	CODE	DEFINITION	A	Accepted This code value can only be used if there are no AK2 loops or all IK501 values = 'A'.	E	Accepted, But Errors Were Noted. The functional group indicated in this 999 contained errors, but was forwarded for further processing.	M	Rejected, Message Authentication Code (MAC) Failed	P	Partially Accepted, At Least One Transaction Set Was Rejected	R	Rejected The functional group indicated in this 999 contained errors, and was NOT forwarded for further processing. It will need to be corrected and resubmitted.	
CODE	DEFINITION															
A	Accepted This code value can only be used if there are no AK2 loops or all IK501 values = 'A'.															
E	Accepted, But Errors Were Noted. The functional group indicated in this 999 contained errors, but was forwarded for further processing.															
M	Rejected, Message Authentication Code (MAC) Failed															
P	Partially Accepted, At Least One Transaction Set Was Rejected															
R	Rejected The functional group indicated in this 999 contained errors, and was NOT forwarded for further processing. It will need to be corrected and resubmitted.															

			W	Rejected, Assurance Failed Validity Tests			
			X	Rejected, Content After Decryption Could Not Be Analyzed			
REQUIRED	AK902	97		Number of Transaction Sets Included	M 1	N0	1/6
				Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element			
REQUIRED	AK903	123		Number of Received Transaction Sets	M 1	N0	1/6
				Number of Transaction Sets received			
REQUIRED	AK904	2		Number of Accepted Transaction Sets	M 1	N0	1/6
				Number of accepted Transaction Sets in a Functional Group			
SITUATIONAL	AK905	716		Functional Group Syntax Error Code	O 1	ID	1/3
				Code indicating error found based on the syntax editing of the functional group header and/or trailer			

SITUATIONAL RULE: *Required when AK901 = E or R, and the error is at the functional group level. If not required by this implementation guide, do not send.*

CODE	DEFINITION
1	Functional Group Not Supported
2	Functional Group Version Not Supported
3	Functional Group Trailer Missing
4	Group Control Number in the Functional Group Header and Trailer Do Not Agree
5	Number of Included Transaction Sets Does Not Match Actual Count
6	Group Control Number Violates Syntax
10	Authentication Key Name Unknown
11	Encryption Key Name Unknown
12	Requested Service (Authentication or Encryption) Not Available
13	Unknown Security Recipient
14	Unknown Security Originator
15	Syntax Error in Decrypted Text
16	Security Not Supported
17	Incorrect Message Length (Encryption Only)
18	Message Authentication Code Failed
19	Functional Group Control Number not Unique within Interchange
23	S3E Security End Segment Missing for S3S Security Start Segment
24	S3S Security Start Segment Missing for S3E End Segment
25	S4E Security End Segment Missing for S4S Security Start Segment
26	S4S Security Start Segment Missing for S4E Security End Segment

SITUATIONAL	AK906	716	Functional Group Syntax Error Code Code indicating error found based on the syntax editing of the functional group header and/or trailer SITUATIONAL RULE: <i>Required when AK901 = E or R, and AK905 has been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3
SITUATIONAL	AK907	716	Functional Group Syntax Error Code Code indicating error found based on the syntax editing of the functional group header and/or trailer SITUATIONAL RULE: <i>Required when AK901 = E or R, and AK905 and AK906 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3
SITUATIONAL	AK908	716	Functional Group Syntax Error Code Code indicating error found based on the syntax editing of the functional group header and/or trailer SITUATIONAL RULE: <i>Required when AK901 = E or R, and AK905, AK906, and AK907 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3
SITUATIONAL	AK909	716	Functional Group Syntax Error Code Code indicating error found based on the syntax editing of the functional group header and/or trailer SITUATIONAL RULE: <i>Required when AK901 = E or R, and AK905, AK906, AK907, and AK908 have been used, and there are additional error codes to report. If not required by this implementation guide, do not send.</i>	O 1	ID	1/3

SEGMENT DETAIL

SE - TRANSACTION SET TRAILER

X12 Segment Name: Transaction Set Trailer

X12 Purpose: To indicate the end of the transaction set and provide the count of the transmitted segments (including the beginning (ST) and ending (SE) segments)

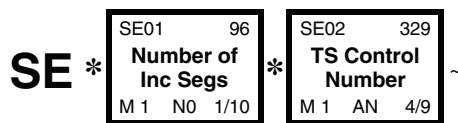
X12 Comments: 1. SE is the last segment of each transaction set.

Segment Repeat: 1

Usage: REQUIRED

TR3 Example: SE*53*0001~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	SE01	96	Number of Included Segments Total number of segments included in a transaction set including ST and SE segments	M 1 NO 1/10
REQUIRED	SE02	329	Transaction Set Control Number Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set	M 1 AN 4/9

The Transaction Set Control Numbers in ST02 and SE02 must be identical. The number is assigned by the originator and must be unique within a functional group (GS-GE). The number also aids in error resolution research. For example, start with the number 0001 and increment from there.

3 Example

3.1 EDI Transmission Example

The following example describes a 999 transaction set that is responding to a functional group that was received containing three 837 transaction sets. The first transaction set conformed fully with the X12 standard, while the second and third contained errors.

The Interchange Control and Functional Group segments (ISA, GS, GE, and IEA) are required in the ASC X12 message. See Appendix C for additional details on these segments.

```
ISA*00*                *00*                *ZZ*123456789
*ZZ*987654321
*041117*1024**^*00501*000000286*0*P*:~
```

```
GS*FA*RCVR*SNDR*20041117*1024*287*X*005010X231A1~
```

The ST segment indicates the beginning of the 999 transaction set, control number 2870001.

```
ST*999*2870001*005010X231A1~
```

The AK1 segment describes the functional group to which this 999 is responding.

```
AK1*HC*17456*004010X098A1~
```

The first Transaction Response Loop indicates that the received transaction set, control number 0001, was accepted with no errors.

```
AK2*837*0001~
IK5*A~
```

The second Transaction Response Loop indicates that the received transaction set, control number 0002, was rejected due to a missing CLM02 data element. The CTX segment identifies the Business Unit (i.e. the claim) that was in error.

```
AK2*837*0002~
IK3*CLM*22**8~
CTX*CLM01:123456789~
IK4*2*782*1~
IK5*R*5~
```

The third Transaction Response Loop indicates that the received transaction set, control number 003, was rejected due to a missing REF (Original Reference Number ICN/DCN) segment. This segment is required by the implementation guide when CLM05-3 = 6, 7, or 8. The IK3 segment indicates the missing REF segment, and the first CTX segment indicates the CLM05-3 as the reason for the missing REF segment. The second CTX identifies the Business Unit (i.e. the claim) that was in error.

```
AK2*837*0003~
IK3*REF*57**3~
CTX*SITUATIONAL TRIGGER*CLM*43**5:3~
```

CTX*CLM01:987654321~

IK5*R*5~

The Trailer section provides a summary of the disposition of the received functional group, and ends the transaction set.

AK9*P*3*3*1~

SE*16*2870001~

GE*1*287~

IEA*1*000000286~

A External Code Sources

77 X12 Directories

SIMPLE DATA ELEMENT/CODE REFERENCES

721, 725

SOURCE

X12.3 Data Element Dictionary

X12.22 Segment Directory

AVAILABLE FROM

Data Interchange Standards Association, Inc. (DISA)

Suite 200

1800 Diagonal Road

Alexandria, VA 22314-2852

ABSTRACT

The data element dictionary contains the format and descriptions of data elements used to construct X12 segments. It also contains code lists associated with these data elements. The segment directory contains the format and definitions of the data segments used to construct X12 transaction sets.

881 Version / Release / Industry Identifier Code

SIMPLE DATA ELEMENT/CODE REFERENCES

480

SOURCE

Data Interchange Standards Association

AVAILABLE FROM

Data Interchange Standards Association

333 John Carlyle Street, Suite 600

Alexandria, VA 22314

ABSTRACT

Code indicating the version, release, sub-release, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and sub-release, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed.

B Nomenclature

B.1 ASC X12 Nomenclature

B.1.1 Interchange and Application Control Structures

Appendix B is provided as a reference to the X12 syntax, usage, and related information. It is not a full statement of Interchange and Control Structure rules. The full X12 Interchange and Control Structures and other rules (X12.5, X12.6, X12.59, X12 dictionaries, other X12 standards and official documents) apply unless specifically modified in the detailed instructions of this implementation guide (see Section B.1.1.3.1.2 for an example of such a modification).

B.1.1.1 Interchange Control Structure

The transmission of data proceeds according to very strict format rules to ensure the integrity and maintain the efficiency of the interchange. Each business grouping of data is called a transaction set. For instance, a group of benefit enrollments sent from a sponsor to a payer is considered a transaction set.

Each transaction set contains groups of logically related data in units called segments. For instance, the N4 segment used in the transaction set conveys the city, state, ZIP Code, and other geographic information. A transaction set contains multiple segments, so the addresses of the different parties, for example, can be conveyed from one computer to the other. An analogy would be that the transaction set is like a freight train; the segments are like the train's cars; and each segment can contain several data elements the same as a train car can hold multiple crates.

The sequence of the elements within one segment is specified by the ASC X12 standard as well as the sequence of segments in the transaction set. In a more conventional computing environ-

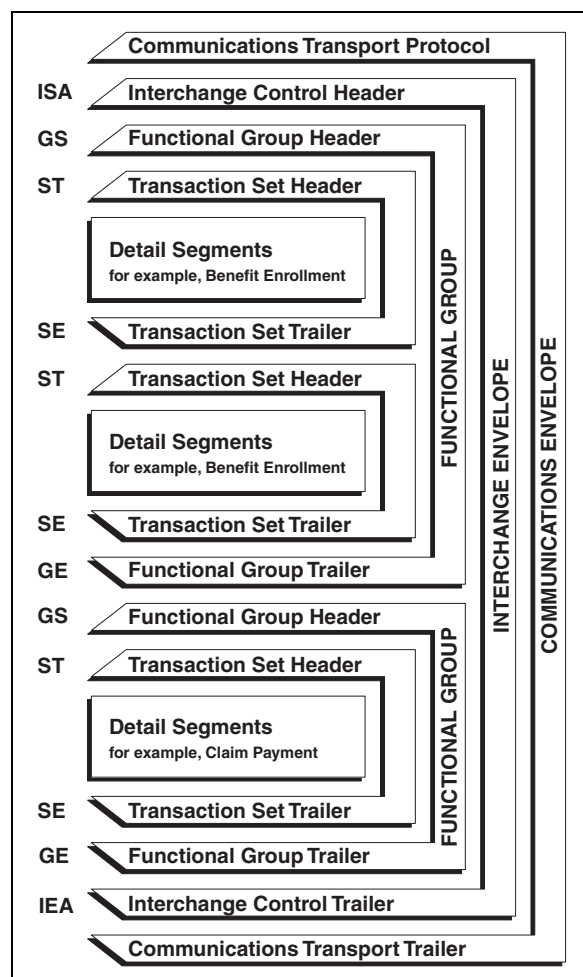


Figure B.1. Transmission Control Schematic

ment, the segments would be equivalent to records, and the elements equivalent to fields.

Similar transaction sets, called “functional groups,” can be sent together within a transmission. Each functional group is prefaced by a group start segment; and a functional group is terminated by a group end segment. One or more functional groups are prefaced by an interchange header and followed by an interchange trailer. Figure B.1., Transmission Control Schematic, illustrates this interchange control.

The interchange header and trailer segments envelop one or more functional groups or interchange-related control segments and perform the following functions:

1. Define the data element separators and the data segment terminator.
2. Identify the sender and receiver.
3. Provide control information for the interchange.
4. Allow for authorization and security information.

B.1.1.2 Application Control Structure Definitions and Concepts

B.1.1.2.1 Basic Structure

A data element corresponds to a data field in data processing terminology. A data segment corresponds to a record in data processing terminology. The data segment begins with a segment ID and contains related data elements. A control segment has the same structure as a data segment; the distinction is in the use. The data segment is used primarily to convey user information, but the control segment is used primarily to convey control information and to group data segments.

B.1.1.2.2 Basic Character Set

The section that follows is designed to have representation in the common character code schemes of EBCDIC, ASCII, and CCITT International Alphabet 5. The ASC X12 standards are graphic-character-oriented; therefore, common character encoding schemes other than those specified herein may be used as long as a common mapping is available. Because the graphic characters have an implied mapping across character code schemes, those bit patterns are not provided here.

The basic character set of this standard, shown in Figure B.2., Basic Character Set, includes those selected from the uppercase letters, digits, space, and special characters as specified below.

A...Z	0...9	!	“	&	'	(	)	*	+
,	-	.	/	:	;	?	=	“ ” (space)	

Figure B.2. Basic Character Set

B.1.1.2.3**Extended Character Set**

An extended character set may be used by negotiation between the two parties and includes the lowercase letters and other special characters as specified in Figure B.3., Extended Character Set.

a..z	%	~	@	[	]	_	{
}	\		<	>	#	\$	

Figure B.3. Extended Character Set

Note that the extended characters include several character codes that have multiple graphical representations for a specific bit pattern. The complete list appears in other standards such as CCITT S.5. Use of the USA graphics for these codes presents no problem unless data is exchanged with an international partner. Other problems, such as the translation of item descriptions from English to French, arise when exchanging data with an international partner, but minimizing the use of codes with multiple graphics eliminates one of the more obvious problems.

For implementations compliant with this guide, either the entire extended character set must be acceptable, or the entire extended character set must not be used. In the absence of a specific trading partner agreement to the contrary, trading partners will assume that the extended character set is acceptable. Use of the extended character set allows the use of the “@” character in email addresses within the PER segment. Users should note that characters in the extended character set, as well as the basic character set, may be used as delimiters only when they do not occur in the data as stated in Section B.1.1.2.5.

B.1.1.2.4**Control Characters**

Two control character groups are specified; they have restricted usage. The common notation for these groups is also provided, together with the character coding in three common alphabets. In the Matrix B.1., Base Control Set, the column IA5 represents CCITT V.3 International Alphabet 5.

B.1.1.2.4.1**Base Control Set**

The base control set includes those characters that will not have a disruptive effect on most communication protocols. These are represented by:

NOTATION	NAME	EBCDIC	ASCII	IA5
BEL	bell	2F	07	07
HT	horizontal tab	05	09	09
LF	line feed	25	0A	0A
VT	vertical tab	0B	0B	0B
FF	form feed	0C	0C	0C
CR	carriage return	0D	0D	0D
FS	file separator	1C	1C	1C
GS	group separator	1D	1D	1D
RS	record separator	1E	1E	1E
US	unit separator	1F	1F	1F
NL	new line	15		

Matrix B.1. Base Control Set

The Group Separator (GS) may be an exception in this set because it is used in the 3780 communications protocol to indicate blank space compression.

B.1.1.2.4.2**Extended Control Set**

The extended control set includes those that may have an effect on a transmission system. These are shown in Matrix B.2., Extended Control Set.

NOTATION	NAME	EBCDIC	ASCII	IA5
SOH	start of header	01	01	01
STX	start of text	02	02	02
ETX	end of text	03	03	03
EOT	end of transmission	37	04	04
ENQ	enquiry	2D	05	05
ACK	acknowledge	2E	06	06
DC1	device control 1	11	11	11
DC2	device control 2	12	12	12
DC3	device control 3	13	13	13
DC4	device control 4	3C	14	14
NAK	negative acknowledge	3D	15	15
SYN	synchronous idle	32	16	16
ETB	end of block	26	17	17

Matrix B.2. Extended Control Set

B.1.1.2.5**Delimiters**

A delimiter is a character used to separate two data elements or component elements or to terminate a segment. The delimiters are an integral part of the data.

Delimiters are specified in the interchange header segment, ISA. The ISA segment can be considered in implementations compliant with this guide (see Appendix C, ISA Segment Note 1) to be a 105 byte fixed length record, followed by a segment terminator. The data element separator is byte number 4; the repetition separator is byte number 83; the component element separator is byte number 105; and the segment terminator is the byte that immediately follows the component element separator.

Once specified in the interchange header, the delimiters are not to be used in a data element value elsewhere in the interchange. For consistency, this implementation guide uses the delimiters shown in Matrix B.3., Delimiters, in all examples of EDI transmissions.

CHARACTER	NAME	DELIMITER
*	Asterisk	Data Element Separator
^	Caret	Repetition Separator
:	Colon	Component Element Separator
~	Tilde	Segment Terminator

Matrix B.3. Delimiters

The delimiters above are for illustration purposes only and are not specific recommendations or requirements. Users of this implementation guide should be aware that an application system may use some valid delimiter characters within the application data. Occurrences of delimiter characters in transmitted data within a data element will result in errors in translation. The existence of asterisks (*) within transmitted application data is a known issue that can affect translation software.

B.1.1.3**Business Transaction Structure Definitions and Concepts**

The ASC X12 standards define commonly used business transactions (such as a health care claim) in a formal structure called “transaction sets.” A transaction set is composed of a transaction set header control segment, one or more data segments, and a transaction set trailer control segment. Each segment is composed of the following:

- A unique segment ID
- One or more logically related data elements each preceded by a data element separator
- A segment terminator

B.1.1.3.1**Data Element**

The data element is the smallest named unit of information in the ASC X12 standard. Data elements are identified as either simple or component. A data element that occurs as an ordinal member of a composite data structure is identified as a component data element. A data element that occurs in a segment outside the defined boundaries of a composite data structure is identified as a simple data element. The distinction between simple and component data elements is strictly a matter of context because a data element can be used in either capacity.

Data elements are assigned a unique reference number. Each data element has a name, description, type, minimum length, and maximum length. For ID type data elements, this guide provides the applicable ASC X12 code values and their descriptions or references where the valid code list can be obtained.

A simple data element within a segment may have an attribute indicating that it may occur once or a specific number of times more than once. The number of permitted repeats are defined as an attribute in the individual segment where the repeated data element occurs.

Each data element is assigned a minimum and maximum length. The length of the data element value is the number of character positions used except as noted for numeric, decimal, and binary elements.

The data element types shown in Matrix B.4., Data Element Types, appear in this implementation guide.

SYMBOL	TYPE
Nn	Numeric
R	Decimal
ID	Identifier
AN	String
DT	Date
TM	Time
B	Binary

Matrix B.4. Data Element Types

The data element minimum and maximum lengths may be restricted in this implementation guide for a compliant implementation. Such restrictions may occur by virtue of the allowed qualifier for the data element or by specific instructions regarding length or format as stated in this implementation guide.

B.1.1.3.1.1**Numeric**

A numeric data element is represented by one or more digits with an optional leading sign representing a value in the normal base of 10. The value of a numeric data element includes an implied decimal point. It is used when the position of the decimal point within the data is permanently fixed and is not to be transmitted with the data.

This set of guides denotes the number of implied decimal positions. The representation for this data element type is “Nn” where N indicates that it is numeric and n indicates the number of decimal positions to the right of the implied decimal point.

If n is 0, it need not appear in the specification; N is equivalent to N0. For negative values, the leading minus sign (-) is used. Absence of a sign indicates a positive value. The plus sign (+) must not be transmitted.

EXAMPLE

A transmitted value of 1234, when specified as numeric type N2, represents a value of 12.34.

Leading zeros must be suppressed unless necessary to satisfy a minimum length requirement. The length of a numeric type data element does not include the optional sign.

B.1.1.3.1.2**Decimal**

A decimal data element may contain an explicit decimal point and is used for numeric values that have a varying number of decimal positions. This data element type is represented as “R.”

The decimal point always appears in the character stream if the decimal point is at any place other than the right end. If the value is an integer (decimal point at the right end) the decimal point must be omitted. For negative values, the leading minus sign (-) is used. Absence of a sign indicates a positive value. The plus sign (+) must not be transmitted.

Leading zeros must be suppressed unless necessary to satisfy a minimum length requirement. Trailing zeros following the decimal point must be suppressed unless necessary to indicate precision. The use of triad separators (for example, the commas in 1,000,000) is expressly prohibited. The length of a decimal type data element does not include the optional leading sign or decimal point.

EXAMPLE

A transmitted value of 12.34 represents a decimal value of 12.34.

While the ASC X12 standard supports usage of exponential notation, this guide prohibits that usage.

For implementation of this guide under the rules promulgated under the Health Insurance Portability and Accountability Act (HIPAA), decimal data elements in Data Element 782 (Monetary Amount) will be limited to a maximum length of 10 characters including reported or implied places for cents (implied value of 00 after the decimal point). Note the statement in the preceding paragraph that the decimal point and leading sign, if sent, are not part of the character count.

EXAMPLE

For implementations mandated under HIPAA rules:

- The following transmitted value represents the largest positive dollar amount that can be sent:
99999999.99
- The following transmitted value is the longest string of characters that can be sent representing whole dollars.
99999999
- The following transmitted value is the longest string of characters that can be sent representing negative dollars and cents.
-99999999.99
- The following transmitted value is the longest string of characters that can be sent representing negative whole dollars.
-99999999

B.1.1.3.1.3**Identifier**

An identifier data element always contains a value from a predefined list of codes that is maintained by the ASC X12 Committee or some other body recognized by the Committee. Trailing spaces must be suppressed unless they are necessary to satisfy a minimum length. An identifier is always left justified. The representation for this data element type is "ID."

B.1.1.3.1.4**String**

A string data element is a sequence of any characters from the basic or extended character sets. The string data element must contain at least one non-space character. The significant characters shall be left justified. Leading spaces, when they occur, are presumed to be significant characters. Trailing spaces must be suppressed unless they are necessary to satisfy a minimum length. The representation for this data element type is "AN."

B.1.1.3.1.5**Date**

A date data element is used to express the standard date in either YYMMDD or CCYYMMDD format in which CC is the first two digits of the calendar year, YY is the last two digits of the calendar year, MM is the month (01 to 12), and DD is the day in the month (01 to 31). The representation for this data element type is "DT." Users of this guide should note that all dates within transactions are 8-character dates (millennium compliant) in the format CCYYMMDD. The only date data element that is in format YYMMDD is the Interchange Date data element in the ISA segment and the TA1 segment where the century is easily determined because of the nature of an interchange header.

B.1.1.3.1.6**Time**

A time data element is used to express the ISO standard time HHMMSSd..d format in which HH is the hour for a 24 hour clock (00 to 23), MM is the minute (00 to 59), SS is the second (00 to 59) and d..d is decimal seconds. The representation for this data element type is "TM." The length of the data element determines the format of the transmitted time.

EXAMPLE

Transmitted data elements of four characters denote HHMM. Transmitted data elements of six characters denote HHMMSS.

B.1.1.3.1.7**Binary**

The binary data element is any sequence of octets ranging in value from binary 00000000 to binary 11111111. This data element type has no defined maximum length. Actual length is specified by the immediately preceding data element. Within the body of a transaction set (from ST to SE) implemented according to this technical report, the binary data element type is only used in the segments Binary Data Segment BIN, and Binary Data Structure BDS. Within those segments, Data Element 785 Binary Data is a string of octets which can assume any binary pattern from hexadecimal 00 to FF, and can be used to send text as well as coded data, including data from another application in its native format. The binary data type is also used in some control and security structures.

Not all transaction sets use the Binary Data Segment BIN or Binary Data Structure BDS.

B.1.1.3.2**Repeating Data Elements**

Simple or composite data elements within a segment can be designated as repeating data elements. Repeating data elements are adjacent data elements that occur up to a number of times specified in the standard as number of repeats. The implementation guide may also specify the number of repeats of a repeating data element in a specific location in the transaction that are permitted in a compliant implementation. Adjacent occurrences of the same repeating simple data element or composite data structure in a segment shall be separated by a repetition separator.

B.1.1.3.3**Composite Data Structure**

The composite data structure is an intermediate unit of information in a segment. Composite data structures are composed of one or more logically related simple data elements, each, except the last, followed by a sub-element separator. The final data element is followed by the next data element separator or the segment terminator. Each simple data element within a composite is called a component.

Each composite data structure has a unique four-character identifier, a name, and a purpose. The identifier serves as a label for the composite. A composite data structure can be further defined through the use of syntax notes, semantic notes, and comments. Each component within the composite is further characterized by a reference designator and a condition designator. The reference designators and the condition designators are described in Sections B.1.1.3.8 and B.1.1.3.9.

A composite data structure within a segment may have an attribute indicating that it may occur once or a specific number of times more than once. The number of permitted repeats are defined as an attribute in the individual segment where the repeated composite data structure occurs.

B.1.1.3.4**Data Segment**

The data segment is an intermediate unit of information in a transaction set. In the data stream, a data segment consists of a segment identifier, one or more composite data structures or simple data elements each preceded by a data element separator and succeeded by a segment terminator.

Each data segment has a unique two- or three-character identifier, a name, and a purpose. The identifier serves as a label for the data segment. A segment can be further defined through the use of syntax notes, semantic notes, and comments. Each simple data element or composite data structure within the segment is further characterized by a reference designator and a condition designator.

B.1.1.3.5**Syntax Notes**

Syntax notes describe relational conditions among two or more data segment units within the same segment, or among two or more component data elements within the same composite data structure. For a complete description of the relational conditions, See B.1.1.3.9, Condition Designator.

B.1.1.3.6**Semantic Notes**

Simple data elements or composite data structures may be referenced by a semantic note within a particular segment. A semantic note provides important additional information regarding the intended meaning of a designated data element, particularly a generic type, in the context of its use within a specific data segment. Semantic notes may also define a relational condition among data elements in a segment based on the presence of a specific value (or one of a set of values) in one of the data elements.

B.1.1.3.7**Comments**

A segment comment provides additional information regarding the intended use of the segment.

B.1.1.3.8**Reference Designator**

Each simple data element or composite data structure in a segment is provided a structured code that indicates the segment in which it is used and the sequential position within the segment. The code is composed of the segment identifier followed by a two-digit number that defines the position of the simple data element or composite data structure in that segment.

For purposes of creating reference designators, the composite data structure is viewed as the hierarchical equal of the simple data element. Each component data element in a composite data structure is identified by a suffix appended to the reference designator for the composite data structure of which it is a member. This suffix is a two-digit number, prefixed with a hyphen, that defines the position of the component data element in the composite data structure.

EXAMPLE

- The first simple element of the CLP segment would be identified as CLP01.
- The first position in the SVC segment is occupied by a composite data structure that contains seven component data elements, the reference designator for the second component data element would be SVC01-02.

B.1.1.3.9**Condition Designator**

This section provides information about X12 standard conditions designators. It is provided so that users will have information about the general standard. Implementation guides may impose other conditions designators. See implementation guide section 2.1 Presentation Examples for detailed information about the implementation guide Industry Usage requirements for compliant implementation.

Data element conditions are of three types: mandatory, optional, and relational. They define the circumstances under which a data element may be required to be present or not present in a particular segment.

DESIGNATOR	DESCRIPTION												
M- Mandatory	The designation of mandatory is absolute in the sense that there is no dependency on other data elements. This designation may apply to either simple data elements or composite data structures. If the designation applies to a composite data structure, then at least one value of a component data element in that composite data structure shall be included in the data segment.												
O- Optional	The designation of optional means that there is no requirement for a simple data element or composite data structure to be present in the segment. The presence of a value for a simple data element or the presence of value for any of the component data elements of a composite data structure is at the option of the sender.												
X- Relational	Relational conditions may exist among two or more simple data elements within the same data segment based on the presence or absence of one of those data elements (presence means a data element must not be empty). Relational conditions are specified by a condition code (see table below) and the reference designators of the affected data elements. A data element may be subject to more than one relational condition. The definitions for each of the condition codes used within syntax notes are detailed below: <table><tr><th>CONDITION CODE</th><th>DEFINITION</th></tr><tr><td>P- Paired or Multiple</td><td>If any element specified in the relational condition is present, then all of the elements specified must be present.</td></tr><tr><td>R- Required</td><td>At least one of the elements specified in the condition must be present.</td></tr><tr><td>E- Exclusion</td><td>Not more than one of the elements specified in the condition may be present.</td></tr><tr><td>C- Conditional</td><td>If the first element specified in the condition is present, then all other elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.</td></tr><tr><td>L- List Conditional</td><td>If the first element specified in the condition is present, then at least one of the remaining elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.</td></tr></table>	CONDITION CODE	DEFINITION	P- Paired or Multiple	If any element specified in the relational condition is present, then all of the elements specified must be present.	R- Required	At least one of the elements specified in the condition must be present.	E- Exclusion	Not more than one of the elements specified in the condition may be present.	C- Conditional	If the first element specified in the condition is present, then all other elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.	L- List Conditional	If the first element specified in the condition is present, then at least one of the remaining elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.
CONDITION CODE	DEFINITION												
P- Paired or Multiple	If any element specified in the relational condition is present, then all of the elements specified must be present.												
R- Required	At least one of the elements specified in the condition must be present.												
E- Exclusion	Not more than one of the elements specified in the condition may be present.												
C- Conditional	If the first element specified in the condition is present, then all other elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.												
L- List Conditional	If the first element specified in the condition is present, then at least one of the remaining elements must be present. However, any or all of the elements not specified as the first element in the condition may appear without requiring that the first element be present. The order of the elements in the condition does not have to be the same as the order of the data elements in the data segment.												

Table B.5. Condition Designator

B.1.1.3.10

Absence of Data

Any simple data element that is indicated as mandatory must not be empty if the segment is used. At least one component data element of a composite data structure that is indicated as mandatory must not be empty if the segment is used. Optional simple data elements and/or composite data structures and their preceding data element separators that are not needed must be omitted if they occur at the end of a segment. If they do not occur at the end of the segment, the simple data element values and/or composite data structure values may be omitted. Their ab-

sence is indicated by the occurrence of their preceding data element separators, in order to maintain the element's or structure's position as defined in the data segment.

Likewise, when additional information is not necessary within a composite, the composite may be terminated by providing the appropriate data element separator or segment terminator.

If a segment has no data in any data element within the segment (an "empty" segment), that segment must not be sent.

B.1.1.3.11**Control Segments**

A control segment has the same structure as a data segment, but it is used for transferring control information rather than application information.

B.1.1.3.11.1**Loop Control Segments**

Loop control segments are used only to delineate bounded loops. Delineation of the loop shall consist of the loop header (LS segment) and the loop trailer (LE segment). The loop header defines the start of a structure that must contain one or more iterations of a loop of data segments and provides the loop identifier for this loop. The loop trailer defines the end of the structure. The LS segment appears only before the first occurrence of the loop, and the LE segment appears only after the last occurrence of the loop. Unbounded looping structures do not use loop control segments.

B.1.1.3.11.2**Transaction Set Control Segments**

The transaction set is delineated by the transaction set header (ST segment) and the transaction set trailer (SE segment). The transaction set header identifies the start and identifier of the transaction set. The transaction set trailer identifies the end of the transaction set and provides a count of the data segments, which includes the ST and SE segments.

B.1.1.3.11.3**Functional Group Control Segments**

The functional group is delineated by the functional group header (GS segment) and the functional group trailer (GE segment). The functional group header starts and identifies one or more related transaction sets and provides a control number and application identification information. The functional group trailer defines the end of the functional group of related transaction sets and provides a count of contained transaction sets.

B.1.1.3.11.4**Relations among Control Segments**

The control segment of this standard must have a nested relationship as is shown and annotated in this subsection. The letters preceding the control segment name are the segment identifier for that control segment. The indentation of segment identifiers shown below indicates the subordination among control segments.

GS Functional Group Header, starts a group of related transaction sets.

ST Transaction Set Header, starts a transaction set.

LS Loop Header, starts a bounded loop of data segments but is not part of the loop.

LS Loop Header, starts an inner, nested, bounded loop.

LE Loop Trailer, ends an inner, nested bounded loop.

LE Loop Trailer, ends a bounded loop of data segments but is not part of the loop.

SE Transaction Set Trailer, ends a transaction set.

GE Functional Group Trailer, ends a group of related transaction sets.

More than one ST/SE pair, each representing a transaction set, may be used within one functional group. Also more than one LS/LE pair, each representing a bounded loop, may be used within one transaction set.

B.1.1.3.12

Transaction Set

The transaction set is the smallest meaningful set of information exchanged between trading partners. The transaction set consists of a transaction set header segment, one or more data segments in a specified order, and a transaction set trailer segment. See Figure B.1., Transmission Control Schematic.

B.1.1.3.12.1

Transaction Set Header and Trailer

A transaction set identifier uniquely identifies a transaction set. This identifier is the first data element of the Transaction Set Header Segment (ST). A user assigned transaction set control number in the header must match the control number in the Trailer Segment (SE) for any given transaction set. The value for the number of included segments in the SE segment is the total number of segments in the transaction set, including the ST and SE segments.

B.1.1.3.12.2

Data Segment Groups

The data segments in a transaction set may be repeated as individual data segments or as unbounded or bounded loops.

B.1.1.3.12.3

Repeated Occurrences of Single Data Segments

When a single data segment is allowed to be repeated, it may have a specified maximum number of occurrences defined at each specified position within a given transaction set standard. Alternatively, a segment may be allowed to repeat an unlimited number of times. The notation for an unlimited number of repetitions is ">1."

B.1.1.3.12.4

Loops of Data Segments

Loops are groups of semantically related segments. Data segment loops may be unbounded or bounded.

B.1.1.3.12.4.1

Unbounded Loops

To establish the iteration of a loop, the first data segment in the loop must appear once and only once in each iteration. Loops may have a specified maximum number of repetitions. Alternatively, the loop may be specified as having an unlimited number of iterations. The notation for an unlimited number of repetitions is ">1."

B.1.1.3.12.4.2

A specified sequence of segments is in the loop. Loops themselves are optional or mandatory. The requirement designator of the beginning segment of a loop indicates whether at least one occurrence of the loop is required. Each appearance of the beginning segment defines an occurrence of the loop.

The requirement designator of any segment within the loop after the beginning segment applies to that segment for each occurrence of the loop. If there is a mandatory requirement designator for any data segment within the loop after the beginning segment, that data segment is mandatory for each occurrence of the loop. If the loop is optional, the mandatory segment only occurs if the loop occurs.

Bounded Loops

The characteristics of unbounded loops described previously also apply to bounded loops. In addition, bounded loops require a Loop Start Segment (LS) to appear before the first occurrence and a Loop End Segment (LE) to appear after the last consecutive occurrence of the loop. If the loop does not occur, the LS and LE segments are suppressed.

B.1.1.3.12.5

Data Segments in a Transaction Set

When data segments are combined to form a transaction set, three characteristics are applied to each data segment: a requirement designator, a position in the transaction set, and a maximum occurrence.

B.1.1.3.12.6

Data Segment Requirement Designators

A data segment, or loop, has one of the following requirement designators for health care and insurance transaction sets, indicating its appearance in the data stream of a transmission. These requirement designators are represented by a single character code.

DESIGNATOR	DESCRIPTION
M- Mandatory	This data segment must be included in the transaction set. (Note that a data segment may be mandatory in a loop of data segments, but the loop itself is optional if the beginning segment of the loop is designated as optional.)
O- Optional	The presence of this data segment is the option of the sending party.

B.1.1.3.12.7

Data Segment Position

The ordinal positions of the segments in a transaction set are explicitly specified for that transaction. Subject to the flexibility provided by the optional requirement designators of the segments, this positioning must be maintained.

B.1.1.3.12.8

Data Segment Occurrence

A data segment may have a maximum occurrence of one, a finite number greater than one, or an unlimited number indicated by ">1."

B.1.1.3.13

Functional Group

A functional group is a group of similar transaction sets that is bounded by a functional group header segment and a functional group trailer segment. The functional identifier defines the group of transactions that may be included within the functional group. The value for the functional group control number in the header and trailer control segments must be identical for any given group. The value for the number of included transaction sets is the total number of transaction sets in the group. See Figure B.1., Transmission Control Schematic.

B.1.1.4**Envelopes and Control Structures****B.1.1.4.1****Interchange Control Structures**

Typically, the term “interchange” connotes the ISA/IEA envelope that is transmitted between trading/business partners. Interchange control is achieved through several “control” components. The interchange control number is contained in data element ISA13 of the ISA segment. The identical control number must also occur in data element 02 of the IEA segment. Most commercial translation software products will verify that these two elements are identical. In most translation software products, if these elements are different the interchange will be “suspended” in error.

There are many other features of the ISA segment that are used for control measures. For instance, the ISA segment contains data elements such as authorization information, security information, sender identification, and receiver identification that can be used for control purposes. These data elements are agreed upon by the trading partners prior to transmission. The interchange date and time data elements as well as the interchange control number within the ISA segment are used for debugging purposes when there is a problem with the transmission or the interchange.

Data Element ISA12, Interchange Control Version Number, indicates the version of the ISA/IEA envelope. GS08 indicates the version of the transaction sets contained within the ISA/IEA envelope. The versions are not required to be the same. An Interchange Acknowledgment can be requested through data element ISA14. The interchange acknowledgement is the TA1 segment. Data element ISA15, Test Indicator, is used between trading partners to indicate that the transmission is in a “test” or “production” mode. Data element ISA16, Subelement Separator, is used by the translator for interpretation of composite data elements.

The ending component of the interchange or ISA/IEA envelope is the IEA segment. Data element IEA01 indicates the number of functional groups that are included within the interchange. In most commercial translation software products, an aggregate count of functional groups is kept while interpreting the interchange. This count is then verified with data element IEA01. If there is a discrepancy, in most commercial products, the interchange is suspended. The other data element in the IEA segment is IEA02 which is referenced above.

See the Appendix C, EDI Control Directory, for a complete detailing of the interchange control header and trailer. The authors recommend that when two transactions with different X12 versions numbers are sent in one interchange control structure (multiple functional groups within one ISA/IEA envelope), the Interchange Control version used should be that of the most recent transaction version included in the envelope. For the transmission of HIPAA transactions with mixed versions, this would be a compliant enveloping structure.

B.1.1.4.2**Functional Groups**

Control structures within the functional group envelope include the functional identifier code in GS01. The Functional Identifier Code is used by the commercial translation software during interpretation of the interchange to determine the different transaction sets that may be included within the functional group. If an inappropriate transaction set is contained within the functional group, most commercial translation software will suspend the functional group within the interchange. The Application Sender's Code in GS02 can be used to identify the sending unit

B.1.1.4.3

of the transmission. The Application Receiver's Code in GS03 can be used to identify the receiving unit of the transmission. The functional group contains a creation date (GS04) and creation time (GS05) for the functional group. The Group Control Number is contained in GS06. These data elements (GS04, GS05, and GS06) can be used for debugging purposes. GS08, Version/Release/Industry Identifier Code is the version/release/sub-release of the transaction sets being transmitted in this functional group.

The Functional Group Control Number in GS06 must be identical to data element 02 of the GE segment. Data element GE01 indicates the number of transaction sets within the functional group. In most commercial translation software products, an aggregate count of the transaction sets is kept while interpreting the functional group. This count is then verified with data element GE01.

See the Appendix C, EDI Control Directory, for a complete detailing of the functional group header and trailer.

HL Structures

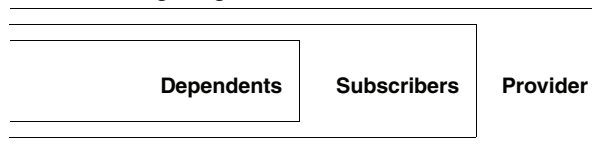
The HL segment is used in several X12 transaction sets to identify levels of detail information using a hierarchical structure, such as relating dependents to a subscriber. Hierarchical levels may differ from guide to guide.

For example, each provider can bill for one or more subscribers, each subscriber can have one or more dependents and the subscriber and the dependents can make one or more claims.

Each guide states what levels are available, the level's usage, number of repeats, and whether that level has subordinate levels within a transaction set.

For implementations compliant with this guide, the repeats of the loops identified by the HL structure shall appear in the hierarchical order specified in BHT01, when those particular hierarchical levels exist. That is, an HL parent loop must be followed by the subordinate child loops, if any, prior to commencing a new HL parent loop at the same hierarchical level.

The following diagram, from transaction set 837, illustrates a typical hierarchy.



The two examples below illustrate this requirement:

Example 1 based on Implementation Guide 811X201:

INSURER

- First STATE in transaction (child of INSURER)
 - First POLICY in transaction (child of first STATE)
 - First VEHICLE in transaction (child of first POLICY)
 - Second POLICY in transaction (child of first STATE)
 - Second VEHICLE in transaction (child of second POLICY)
 - Third VEHICLE in transaction (child of second POLICY)
- Second STATE in transaction (child of INSURER)
 - Third POLICY in transaction (child of second STATE)
 - Fourth VEHICLE in transaction (child of third POLICY)

Example 2 based on Implementation Guide 837X141

First PROVIDER in transaction
 First SUBSCRIBER in transaction (child of first PROVIDER)
Second PROVIDER in transaction
 Second SUBSCRIBER in transaction (child of second PROVIDER)
 First DEPENDENT in transaction (child of second SUBSCRIBER)
 Second DEPENDENT in transaction (child of second SUBSCRIBER)
 Third SUBSCRIBER in transaction (child of second PROVIDER)
Third PROVIDER in transaction
 Fourth SUBSCRIBER in transaction (child of third PROVIDER)
 Fifth SUBSCRIBER in transaction (child of third PROVIDER)
 Third DEPENDENT in transaction (child of fifth SUBSCRIBER)

B.1.1.5**Acknowledgments****B.1.1.5.1****Interchange Acknowledgment, TA1**

The TA1 segment provides the capability for the interchange receiver to notify the sender that a valid envelope was received or that problems were encountered with the interchange control structure. The TA1 verifies the envelopes only. Transaction set-specific verification is accomplished through use of the Functional Acknowledgment Transaction Set, 997. See B.1.1.5.2, Functional Acknowledgment, 997, for more details. The TA1 is unique in that it is a single segment transmitted without the GS/GE envelope structure. A TA1 can be included in an interchange with other functional groups and transactions.

Encompassed in the TA1 are the interchange control number, interchange date and time, interchange acknowledgment code, and the interchange note code. The interchange control number, interchange date and time are identical to those that were present in the transmitted interchange from the trading partner. This provides the capability to associate the TA1 with the transmitted interchange. TA104, Interchange Acknowledgment Code, indicates the status of the interchange control structure. This data element stipulates whether the transmitted interchange was accepted with no errors, accepted with errors, or rejected because of errors. TA105, Interchange Note Code, is a numerical code that indicates the error found while processing the interchange control structure. Values for this data element indicate whether the error occurred at the interchange or functional group envelope.

B.1.1.5.2**Functional Acknowledgment, 997**

The Functional Acknowledgment Transaction Set, 997, has been designed to allow trading partners to establish a comprehensive control function as a part of their business exchange process. This acknowledgment process facilitates control of EDI. There is a one-to-one correspondence between a 997 and a functional group. Segments within the 997 can identify the acceptance or rejection of the functional group, transaction sets or segments. Data elements in error can also be identified. There are many EDI implementations that have incorporated the acknowledgment process in all of their electronic communications. The 997 is used as a functional acknowledgment to a previously transmitted functional group.

The 997 is a transaction set and thus is encapsulated within the interchange control structure (envelopes) for transmission.

B.2 **Object Descriptors**

Object Descriptors (OD) do not apply to this implementation guide.

C EDI Control Directory

C.1 Control Segments

- **ISA**
Interchange Control Header Segment
- **TA1**
Interchange Acknowledgment Segment
- **GS**
Functional Group Header Segment
- **GE**
Functional Group Trailer Segment
- **IEA**
Interchange Control Trailer Segment

SEGMENT DETAIL

ISA - INTERCHANGE CONTROL HEADER

X12 Segment Name: Interchange Control Header

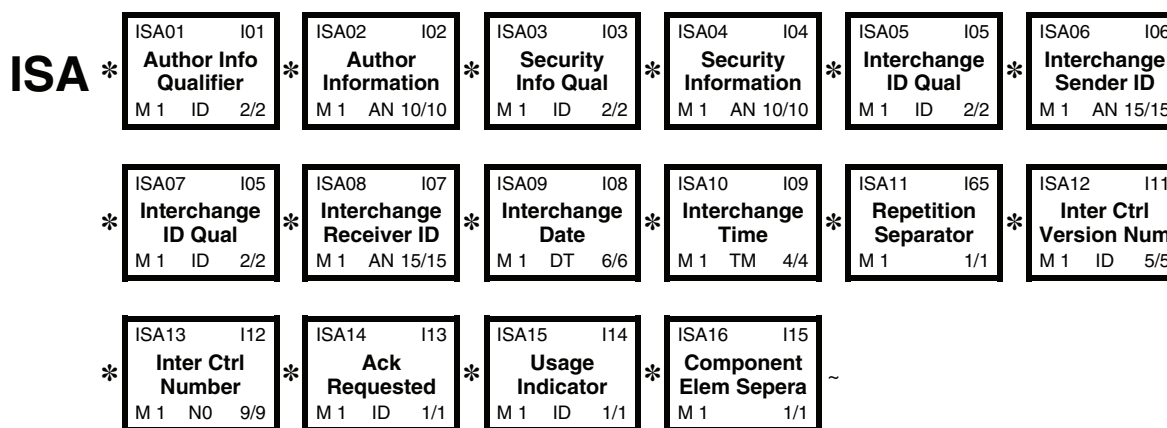
X12 Purpose: To start and identify an interchange of zero or more functional groups and interchange-related control segments

Usage: REQUIRED

- TR3 Notes:**
1. All positions within each of the data elements must be filled.
 2. For compliant implementations under this implementation guide, ISA13, the interchange Control Number, must be a positive unsigned number. Therefore, the ISA segment can be considered a fixed record length segment.
 3. The first element separator defines the element separator to be used through the entire interchange.
 4. The first element separator defines the element separator to be used through the entire interchange.
 5. Spaces in the example interchanges are represented by “.” for clarity.

TR3 Example: ISA*00*.....*01*SECRET....*ZZ*SUBMITTERS.ID..*ZZ*RECEIVERS.ID... *030101*1253*^*00501*000000905*0*T*::~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	ISA01	I01	Authorization Information Qualifier Code identifying the type of information in the Authorization Information	M 1 ID 2/2
			CODE	DEFINITION
			00	No Authorization Information Present (No Meaningful Information in I02)
			01	UCS Communications ID

			02	EDX Communications ID				
			03	Additional Data Identification				
			04	Rail Communications ID				
			05	Department of Defense (DoD) Communication Identifier				
			06	United States Federal Government Communication Identifier				
REQUIRED	ISA02	I02	Authorization Information	M 1	AN	10/10		
			Information used for additional identification or authorization of the interchange sender or the data in the interchange; the type of information is set by the Authorization Information Qualifier (I01)					
REQUIRED	ISA03	I03	Security Information Qualifier	M 1	ID	2/2		
			Code identifying the type of information in the Security Information					
			CODE	DEFINITION				
			00	No Security Information Present (No Meaningful Information in I04)				
			01	Password				
REQUIRED	ISA04	I04	Security Information	M 1	AN	10/10		
			This is used for identifying the security information about the interchange sender or the data in the interchange; the type of information is set by the Security Information Qualifier (I03)					
REQUIRED	ISA05	I05	Interchange ID Qualifier	M 1	ID	2/2		
			Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified					

This ID qualifies the Sender in ISA06.

CODE	DEFINITION
01	Duns (Dun & Bradstreet)
02	SCAC (Standard Carrier Alpha Code)
03	FMC (Federal Maritime Commission)
04	IATA (International Air Transport Association)
07	Global Location Number (GLN)
	CODE SOURCE 583: EAN.UCC Global Location Number (GLN)
08	UCC EDI Communications ID (Comm ID)
09	X.121 (CCITT)
10	Department of Defense (DoD) Activity Address Code
	CODE SOURCE 350: Defense Logistics Management System Manual
11	DEA (Drug Enforcement Administration)
12	Phone (Telephone Companies)
13	UCS Code (The UCS Code is a Code Used for UCS Transmissions; it includes the Area Code and Telephone Number of a Modem; it Does Not Include Punctuation, Blanks or Access Code)
14	Duns Plus Suffix
15	Petroleum Accountants Society of Canada Company Code
16	Duns Number With 4-Character Suffix
17	American Bankers Association (ABA) Transit Routing Number (Including Check Digit, 9 Digit)

18	Association of American Railroads (AAR) Standard Distribution Code CODE SOURCE 420: Association of American Railroads (AAR) Standard Distribution Code
19	EDI Council of Australia (EDICA) Communications ID Number (COMM ID) CODE SOURCE 421: EDI Council of Australia Communications ID Number
20	Health Industry Number (HIN) CODE SOURCE 121: Health Industry Number
21	Integrated Postsecondary Education Data System, or (IPEDS) CODE SOURCE 422: Integrated Postsecondary Education Data System (IPEDS)
22	Federal Interagency Commission on Education, or FICE CODE SOURCE 423: Federal Interagency Commission on Education (FICE)
23	National Center for Education Statistics Common Core of Data 12-Digit Number for Pre-K-Grade 12 Institutes, or NCES CODE SOURCE 424: National Center for Education Statistics (NCES)
24	The College Board's Admission Testing Program 4-Digit Code of Postsecondary Institutes, or ATP CODE SOURCE 425: The College Board's Admission Testing Program (ATP)
25	ACT, Inc. 4-Digit Code of Postsecondary Institutions. CODE SOURCE 426: ACT, Inc.
26	Statistics of Canada List of Postsecondary Institutions CODE SOURCE 296: University Student Information System (USIS) Field of Study Classification CODE SOURCE 300: University Student Information System (USIS) Institution Codes
27	Carrier Identification Number as assigned by Health Care Financing Administration (HCFA)
28	Fiscal Intermediary Identification Number as assigned by Health Care Financing Administration (HCFA)
29	Medicare Provider and Supplier Identification Number as assigned by Health Care Financing Administration (HCFA)
30	U.S. Federal Tax Identification Number
31	Jurisdiction Identification Number Plus 4 as assigned by the International Association of Industrial Accident Boards and Commissions (IAIABC)
32	U.S. Federal Employer Identification Number (FEIN)
33	National Association of Insurance Commissioners Company Code (NAIC)
34	Medicaid Provider and Supplier Identification Number as assigned by individual State Medicaid Agencies in conjunction with Health Care Financing Administration (HCFA)

			35	Statistics Canada Canadian College Student Information System Institution Codes CODE SOURCE 324: Statistics Canada Canadian College Student Information System Institution Codes
			36	Statistics Canada University Student Information System Institution Codes CODE SOURCE 300: University Student Information System (USIS) Institution Codes
			37	Society of Property Information Compilers and Analysts CODE SOURCE 573: Society of Property Information Compilers and Analysts
			38	The College Board and ACT, Inc. 6-Digit Code List of Secondary Institutions CODE SOURCE 862: Secondary School Codes
		AM		Association Mexicana delCodigo de Producto (AMECOP) Communication ID CODE SOURCE 497: AMECOP
		NR		National Retail Merchants Association (NRMA) - Assigned
		SA		User Identification Number as assigned by the Safety and Fitness Electronic Records (SAFER) System CODE SOURCE 851: Safety and Fitness Electronic Records (SAFER) User ID Number
		SN		Standard Address Number CODE SOURCE 42: Book Industry Systems Advisory Committee
		ZZ		Mutually Defined
REQUIRED	ISA06	I06		Interchange Sender ID M 1 AN 15/15 Identification code published by the sender for other parties to use as the receiver ID to route data to them; the sender always codes this value in the sender ID element
REQUIRED	ISA07	I05		Interchange ID Qualifier M 1 ID 2/2 Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified

This ID qualifies the Receiver in ISA08.

CODE	DEFINITION
01	Duns (Dun & Bradstreet)
02	SCAC (Standard Carrier Alpha Code)
03	FMC (Federal Maritime Commission)
04	IATA (International Air Transport Association)
07	Global Location Number (GLN) CODE SOURCE 583: EAN.UCC Global Location Number (GLN)
08	UCC EDI Communications ID (Comm ID)
09	X.121 (CCITT)
10	Department of Defense (DoD) Activity Address Code CODE SOURCE 350: Defense Logistics Management System Manual
11	DEA (Drug Enforcement Administration)
12	Phone (Telephone Companies)

13	UCS Code (The UCS Code is a Code Used for UCS Transmissions; it includes the Area Code and Telephone Number of a Modem; it Does Not Include Punctuation, Blanks or Access Code)
14	Duns Plus Suffix
15	Petroleum Accountants Society of Canada Company Code
16	Duns Number With 4-Character Suffix
17	American Bankers Association (ABA) Transit Routing Number (Including Check Digit, 9 Digit)
18	Association of American Railroads (AAR) Standard Distribution Code CODE SOURCE 420: Association of American Railroads (AAR) Standard Distribution Code
19	EDI Council of Australia (EDICA) Communications ID Number (COMM ID) CODE SOURCE 421: EDI Council of Australia Communications ID Number
20	Health Industry Number (HIN) CODE SOURCE 121: Health Industry Number
21	Integrated Postsecondary Education Data System, or (IPEDS) CODE SOURCE 422: Integrated Postsecondary Education Data System (IPEDS)
22	Federal Interagency Commission on Education, or FICE CODE SOURCE 423: Federal Interagency Commission on Education (FICE)
23	National Center for Education Statistics Common Core of Data 12-Digit Number for Pre-K-Grade 12 Institutes, or NCES CODE SOURCE 424: National Center for Education Statistics (NCES)
24	The College Board's Admission Testing Program 4-Digit Code of Postsecondary Institutes, or ATP CODE SOURCE 425: The College Board's Admission Testing Program (ATP)
25	ACT, Inc. 4-Digit Code of Postsecondary Institutions. CODE SOURCE 426: ACT, Inc.
26	Statistics of Canada List of Postsecondary Institutions CODE SOURCE 296: University Student Information System (USIS) Field of Study Classification CODE SOURCE 300: University Student Information System (USIS) Institution Codes
27	Carrier Identification Number as assigned by Health Care Financing Administration (HCFA)
28	Fiscal Intermediary Identification Number as assigned by Health Care Financing Administration (HCFA)
29	Medicare Provider and Supplier Identification Number as assigned by Health Care Financing Administration (HCFA)
30	U.S. Federal Tax Identification Number

			31	Jurisdiction Identification Number Plus 4 as assigned by the International Association of Industrial Accident Boards and Commissions (IAIABC)			
			32	U.S. Federal Employer Identification Number (FEIN)			
			33	National Association of Insurance Commissioners Company Code (NAIC)			
			34	Medicaid Provider and Supplier Identification Number as assigned by individual State Medicaid Agencies in conjunction with Health Care Financing Administration (HCFA)			
			35	Statistics Canada Canadian College Student Information System Institution Codes CODE SOURCE 324: Statistics Canada Canadian College Student Information System Institution Codes			
			36	Statistics Canada University Student Information System Institution Codes CODE SOURCE 300: University Student Information System (USIS) Institution Codes			
			37	Society of Property Information Compilers and Analysts CODE SOURCE 573: Society of Property Information Compilers and Analysts			
			38	The College Board and ACT, Inc. 6-Digit Code List of Secondary Institutions CODE SOURCE 862: Secondary School Codes			
			AM	Association Mexicana delCodigo de Producto (AMECOP) Communication ID CODE SOURCE 497: AMECOP			
			NR	National Retail Merchants Association (NRMA) - Assigned			
			SA	User Identification Number as assigned by the Safety and Fitness Electronic Records (SAFER) System CODE SOURCE 851: Safety and Fitness Electronic Records (SAFER) User ID Number			
			SN	Standard Address Number CODE SOURCE 42: Book Industry Systems Advisory Committee			
			ZZ	Mutually Defined			
REQUIRED	ISA08	I07		Interchange Receiver ID Identification code published by the receiver of the data; When sending, it is used by the sender as their sending ID, thus other parties sending to them will use this as a receiving ID to route data to them	M 1	AN	15/15
REQUIRED	ISA09	I08		Interchange Date Date of the interchange	M 1	DT	6/6
				The date format is YYMMDD.			
REQUIRED	ISA10	I09		Interchange Time Time of the interchange	M 1	TM	4/4
				The time format is HHMM.			

REQUIRED	ISA11	I65	Repetition Separator	M 1	1/1	Type is not applicable; the repetition separator is a delimiter and not a data element; this field provides the delimiter used to separate repeated occurrences of a simple data element or a composite data structure; this value must be different than the data element separator, component element separator, and the segment terminator
REQUIRED	ISA12	I11	Interchange Control Version Number	M 1 ID	5/5	Code specifying the version number of the interchange control segments
			CODE	DEFINITION		
			00501	Standards Approved for Publication by ASC X12 Procedures Review Board through October 2003		
REQUIRED	ISA13	I12	Interchange Control Number	M 1 N0	9/9	A control number assigned by the interchange sender
			In order to provide sufficient discrimination for the acknowledgment process to operate reliably and to ensure that audit trails are unambiguous, the combination of interchange sender's qualifier and ID (ISA05, ISA06), interchange receiver's qualifier and ID (ISA07, ISA08) and the interchange control number value (ISA13) shall by themselves be unique within a reasonably extended time frame whose boundaries shall be defined by trading partner agreement. Because at some point it may be necessary to reuse a sequence of control numbers, the Interchange Date and Time may serve as an additional discriminant only to differentiate interchange identity over the longest possible time frame.			
REQUIRED	ISA14	I13	Acknowledgment Requested	M 1 ID	1/1	Code indicating sender's request for an interchange acknowledgment
			See Section B.1.1.5.1 for interchange acknowledgment information.			
REQUIRED	ISA15	I14	Interchange Usage Indicator	M 1 ID	1/1	Code indicating whether data enclosed by this interchange envelope is test, production or information
			CODE	DEFINITION		
			I	Information		
			P	Production Data		
			T	Test Data		
REQUIRED	ISA16	I15	Component Element Separator	M 1	1/1	Type is not applicable; the component element separator is a delimiter and not a data element; this field provides the delimiter used to separate component data elements within a composite data structure; this value must be different than the data element separator and the segment terminator

SEGMENT DETAIL

TA1 - INTERCHANGE ACKNOWLEDGMENT

X12 Segment Name: Interchange Acknowledgment

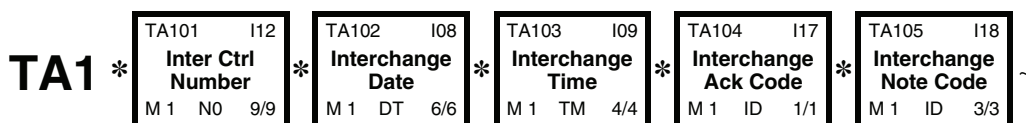
X12 Purpose: To report the status of processing a received interchange header and trailer or the non-delivery by a network provider

Usage: SITUATIONAL

Situational Rule: Required when requested by the sender as indicated in the ISA14 of the submitted interchange. If not required by this implementation guide, do not send.

TR3 Notes: 1. When used, the TA1 segment must either be included in the same interchange as the 999 transaction set, or sent within its own interchange (i.e. ISA-TA1-IEA). When included in the same interchange as the 999 transaction set, the TA1 segment must be included within the interchange (ISA/IEA) and outside of any included functional groups (GS/GE). It is recommended that the TA1 segment be placed between the ISA and first occurrence of the GS segment.

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	TA101	I12	Interchange Control Number A control number assigned by the interchange sender	M 1 NO 9/9
This is the value in ISA13 from the interchange to which this TA1 is responding.				
REQUIRED	TA102	I08	Interchange Date Date of the interchange	M 1 DT 6/6
REQUIRED	TA103	I09	Interchange Time Time of the interchange	M 1 TM 4/4
REQUIRED	TA104	I17	Interchange Acknowledgment Code Code indicating the status of the receipt of the interchange control structure	M 1 ID 1/1
			CODE	DEFINITION
			A	The Transmitted Interchange Control Structure Header and Trailer Have Been Received and Have No Errors.
			E	The Transmitted Interchange Control Structure Header and Trailer Have Been Received and Are Accepted But Errors Are Noted. This Means the Sender Must Not Resend This Data.
			R	The Transmitted Interchange Control Structure Header and Trailer are Rejected Because of Errors.

REQUIRED	TA105	I18	Interchange Note Code	M 1 ID 3/3
			Code specifying the error found processing the interchange control structure	
			CODE	DEFINITION
			000	No error
			001	The Interchange Control Number in the Header and Trailer Do Not Match. The Value From the Header is Used in the Acknowledgment.
			002	This Standard as Noted in the Control Standards Identifier is Not Supported.
			003	This Version of the Controls is Not Supported
			004	The Segment Terminator is Invalid
			005	Invalid Interchange ID Qualifier for Sender
			006	Invalid Interchange Sender ID
			007	Invalid Interchange ID Qualifier for Receiver
			008	Invalid Interchange Receiver ID
			009	Unknown Interchange Receiver ID
			010	Invalid Authorization Information Qualifier Value
			011	Invalid Authorization Information Value
			012	Invalid Security Information Qualifier Value
			013	Invalid Security Information Value
			014	Invalid Interchange Date Value
			015	Invalid Interchange Time Value
			016	Invalid Interchange Standards Identifier Value
			017	Invalid Interchange Version ID Value
			018	Invalid Interchange Control Number Value
			019	Invalid Acknowledgment Requested Value
			020	Invalid Test Indicator Value
			021	Invalid Number of Included Groups Value
			022	Invalid Control Structure
			023	Improper (Premature) End-of-File (Transmission)
			024	Invalid Interchange Content (e.g., Invalid GS Segment)
			025	Duplicate Interchange Control Number
			026	Invalid Data Element Separator
			027	Invalid Component Element Separator
			028	Invalid Delivery Date in Deferred Delivery Request
			029	Invalid Delivery Time in Deferred Delivery Request
			030	Invalid Delivery Time Code in Deferred Delivery Request
			031	Invalid Grade of Service Code

SEGMENT DETAIL

GS - FUNCTIONAL GROUP HEADER

X12 Segment Name: Functional Group Header

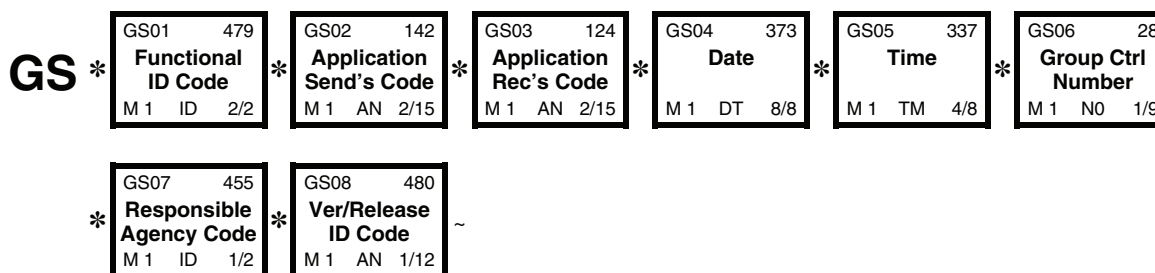
X12 Purpose: To indicate the beginning of a functional group and to provide control information

X12 Comments: 1. A functional group of related transaction sets, within the scope of X12 standards, consists of a collection of similar transaction sets enclosed by a functional group header and a functional group trailer.

Usage: REQUIRED

TR3 Example: GS*FA*SENDER CODE*RECEIVER
CODE*20051231*0802*1*X*005010X231A1~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES						
REQUIRED	GS01	479	Functional Identifier Code Code identifying a group of application related transaction sets	M 1	ID	2/2				
This is the 2-character Functional Identifier Code assigned to each transaction set by X12. The specific code for a transaction set defined by this implementation guide is presented in section 1.2, Version Information.										
The functional identifier code required when creating a 999 transaction set is FA.										
<table><tr><th>CODE</th><th>DEFINITION</th></tr><tr><td>FA</td><td>Functional or Implementation Acknowledgment Transaction Sets (997, 999)</td></tr></table>							CODE	DEFINITION	FA	Functional or Implementation Acknowledgment Transaction Sets (997, 999)
CODE	DEFINITION									
FA	Functional or Implementation Acknowledgment Transaction Sets (997, 999)									
REQUIRED	GS02	142	Application Sender's Code Code identifying party sending transmission; codes agreed to by trading partners	M 1	AN	2/15				
Use this code to identify the unit sending the information.										
REQUIRED	GS03	124	Application Receiver's Code Code identifying party receiving transmission; codes agreed to by trading partners	M 1	AN	2/15				
Use this code to identify the unit receiving the information.										

REQUIRED	GS04	373	Date M 1DT8/8 Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year SEMANTIC: GS04 is the group date. Use this date for the functional group creation date.				
REQUIRED	GS05	337	Time M 1TM4/8 Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99) SEMANTIC: GS05 is the group time. Use this time for the creation time. The recommended format is HHMM.				
REQUIRED	GS06	28	Group Control Number M 1NO1/9 Assigned number originated and maintained by the sender SEMANTIC: The data interchange control number GS06 in this header must be identical to the same data element in the associated functional group trailer, GE02. In order to provide sufficient discrimination for the acknowledgment process to operate reliably and to ensure that audit trails are unambiguous, the combination of Functional ID Code (GS01), Application Sender's ID (GS02), Application Receiver's ID (GS03), and Functional Group Control Numbers (GS06, GE02) shall by themselves be unique within a reasonably extended time frame whose boundaries shall be defined by trading partner agreement. Because at some point it may be necessary to reuse a sequence of control numbers, the Functional Group Date and Time may serve as an additional discriminant only to differentiate functional group identity over the longest possible time frame.				
REQUIRED	GS07	455	Responsible Agency Code M 1ID1/2 Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480 <table><thead><tr><th>CODE</th><th>DEFINITION</th></tr></thead><tbody><tr><td>X</td><td>Accredited Standards Committee X12</td></tr></tbody></table>	CODE	DEFINITION	X	Accredited Standards Committee X12
CODE	DEFINITION						
X	Accredited Standards Committee X12						

REQUIRED	GS08	480	Version / Release / Industry Identifier Code M 1 AN 1/12
			Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed CODE SOURCE 881: Version / Release / Industry Identifier Code This is the unique Version/Release/Industry Identifier Code assigned to an implementation by X12. The specific code for a transaction set defined by this implementation guide is presented in section 1.2, Version Information. Implementation Acknowledgement (999) transactions must use only the Version/Release of the underlying standard used to create the 999.
		CODE	DEFINITION
		005010X231A1	Standards Approved for Publication by ASC X12
			Procedures Review Board through October 2003

SEGMENT DETAIL

GE - FUNCTIONAL GROUP TRAILER

X12 Segment Name: Functional Group Trailer

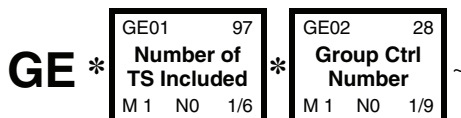
X12 Purpose: To indicate the end of a functional group and to provide control information

X12 Comments: 1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header.

Usage: REQUIRED

TR3 Example: GE*1*1~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES
REQUIRED	GE01	97	Number of Transaction Sets Included Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element	M 1 NO 1/6
REQUIRED	GE02	28	Group Control Number Assigned number originated and maintained by the sender	M 1 NO 1/9

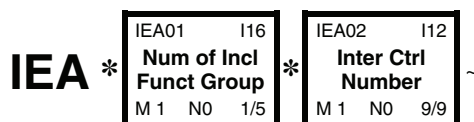
SEMANTIC: The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

The value in GE02 must be identical to the value in GS06.

SEGMENT DETAIL

IEA - INTERCHANGE CONTROL TRAILER**X12 Segment Name:** Interchange Control Trailer**X12 Purpose:** To define the end of an interchange of zero or more functional groups and interchange-related control segments**Usage:** REQUIRED**TR3 Example:** IEA*1*000000905~

DIAGRAM



ELEMENT DETAIL

USAGE	REF. DES.	DATA ELEMENT	NAME	ATTRIBUTES		
REQUIRED	IEA01	I16	Number of Included Functional Groups A count of the number of functional groups included in an interchange	M 1	N0	1/5
REQUIRED	IEA02	I12	Interchange Control Number A control number assigned by the interchange sender	M 1	N0	9/9
The value in IEA02 must be identical to the value in ISA13.						

D Change Summary

This is the first ASC X12N Implementation Guide for the Implementation Acknowledgment business use of the 999. In future guides, this section will contain a summary and detail of all changes since the previous guide.

E Data Element Glossary

E.1 Data Element Name Index

This section contains an alphabetic listing of data elements used in this implementation guide. Consult the X12N Data Element Dictionary for a complete list of all X12N Data Elements. Data element names in normal type are generic ASC X12 names. Italic type indicates a health care industry defined name.

Name	<i>Payment Date</i>
Definition	Date of payment.
Transaction Set ID	277
Locator Key	D 2200D SPA12 C001-2 373 156
H=Header, D=Detail, S=Summary	
Loop ID	
Segment ID/Reference Designator	
Composite ID-Sequence	
Data Element Number	
Page Number	

Component Data Element Position in Composite

To identify the component data element position within the composite that is in error.

H 2100 - CTX05 C030-2 1528 30
AK2/IK3
H 2110 - IK401 C030-2 1528 34
AK2/IK3/I
K4
H 2110 - CTX05 C030-2 1528 37
AK2/IK3/I
K4

Component Data Element Reference Number

Reference number used to locate the data element in the X12 Data Element Dictionary.

H 2100 - CTX06 C999-2 1725 31
AK2/IK3

Context Name

Holds the name or 'tag' of a context.

H 2100 - CTX01 C998-1 1999 29
AK2/IK3
H 2100 - CTX01 C998-1 1999 33
AK2/IK3
H 2110 - CTX01 C998-1 1999 36
AK2/IK3/I
K4

Context Reference

Holds a reference to or for a context.

H 2100 - CTX01 C998-2 1998 33
AK2/IK3

Copy of Bad Data Element

This is a copy of the data element in error.

H 2110 - IK404 - 1724 35
AK2/IK3/I
K4

Data Element Reference Number

Reference number used to locate the data element in the Data Element Dictionary.

H 2100 - CTX06 C999-1 1725 31
AK2/IK3
H 2110 - IK402 - 1725 35
AK2/IK3/I
K4
H 2110 - CTX06 C999-1 1725 38
AK2/IK3/I
K4
H 2110 - CTX06 C999-2 1725 38
AK2/IK3/I
K4

Element Position in Segment

This is used to indicate the relative position of a simple data element or the relative position of a composite data structure with the relative position of the component within the composite data structure.

H 2100 - CTX05 C030-1 1722 30
AK2/IK3
H 2110 - IK401 C030-1 1722 34
AK2/IK3/I
K4
H 2110 - CTX05 C030-1 1722 37
AK2/IK3/I
K4

Functional Group Acknowledge Code

Code indicating accept or reject condition based on the syntax editing of the functional group.

H | | AK901 | - | 1715 42

Functional Group Syntax Error Code

Code indicating error found based on the syntax editing of the functional group header and/or trailer.

H | | AK905 | - | 1716 43

H | | AK906 | - | 1716 44

H | | AK907 | - | 1716 44

H | | AK908 | - | 1716 44

H | | AK909 | - | 1716 44

Functional Identifier Code

Code identifying a group of application related transaction sets.

H | | AK101 | - | 1479 23

Group Control Number

Assigned number originated and maintained by the sender.

H | | AK102 | - | 128 23

Implementation Convention Reference

Reference assigned to identify Implementation Convention.

H | | ST03 | - | 1705 22

H | 2000 - | AK203 | - | 1705 26
AK2

Implementation Data Element Syntax Error Code

Code indicating the implementation error found after syntax edits of a data element.

H | 2110 - | IK403 | - | 1621 35

AK2/IK3/I

K4

Implementation Segment Syntax Error Code

Code indicating implementation error found based on the syntax editing of a segment.

H | 2100 - | IK304 | - | 1620 28

AK2/IK3

Implementation Transaction Set Syntax Error Code

Code indicating implementation error found based on the syntax editing of a transaction set.

H | 2000 - | IK502 | - | 1618 40

AK2

H | 2000 - | IK503 | - | 1618 40

AK2

H | 2000 - | IK504 | - | 1618 41

AK2

H | 2000 - | IK505 | - | 1618 41

AK2

H | 2000 - | IK506 | - | 1618 41

AK2

Loop Identifier Code

The loop ID number given on the transaction set diagram is the value for this data element in segments LS and LE.

H | 2100 - | IK303 | - | 1447 28

AK2/IK3

H | 2100 - | CTX04 | - | 1447 30

AK2/IK3

H | 2110 - | CTX04 | - | 1447 37

AK2/IK3/I

K4

Number of Accepted Transaction Sets

Number of accepted Transaction Sets in a Functional Group.

H | | AK904 | - | 12 43

Number of Included Segments

Total number of segments included in a transaction set including ST and SE segments.

H | | SE01 | - | 196 45

Number of Received Transaction Sets

Number of Transaction Sets received.

H | | AK903 | - | 123 43

Number of Transaction Sets Included

Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element.

H | | AK902 | - | 197 43

Repeating Data Element Position

To identify the specific repetition of a data element that is in error.

H | 2100 - | CTX05 | C030-3 | 1686 30

AK2/IK3

H | 2110 - | IK401 | C030-3 | 1686 35

AK2/IK3/I

K4

H | 2110 - | CTX05 | C030-3 | 1686 37

AK2/IK3/I

K4

Segment ID Code

Code defining the segment ID of the data segment in error (See Appendix A - Number 77).

H		2100 -		IK301		-		721		27
				AK2/IK3						
H		2100 -		CTX02		-		721		29
				AK2/IK3						
H		2110 -		CTX02		-		721		36
				AK2/IK3/I						
				K4						

Segment Position in Transaction Set

The numerical count position of this data segment from the start of the transaction set: the transaction set header is count position 1.

H		2100 -		IK302		-		719		27
				AK2/IK3						
H		2100 -		CTX03		-		719		29
				AK2/IK3						
H		2110 -		CTX03		-		719		36
				AK2/IK3/I						
				K4						

Transaction Set Acknowledgment Code

Code indicating accept or reject condition based on the syntax editing of the transaction set.

H		2000 -		IK501		-		717		39
				AK2						

Transaction Set Control Number

The unique identification number within a transaction set.

H				ST02		-		329		21
H		2000 -		AK202		-		329		26
				AK2						
H				SE02		-		329		45

Transaction Set Identifier Code

Code uniquely identifying a Transaction Set.

H				ST01		-		143		21
H		2000 -		AK201		-		143		25
				AK2						

Version, Release, or Industry Identifier Code

Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used.

H				AK103		-		1480		24
---	--	--	--	-------	--	---	--	------	-------	----

